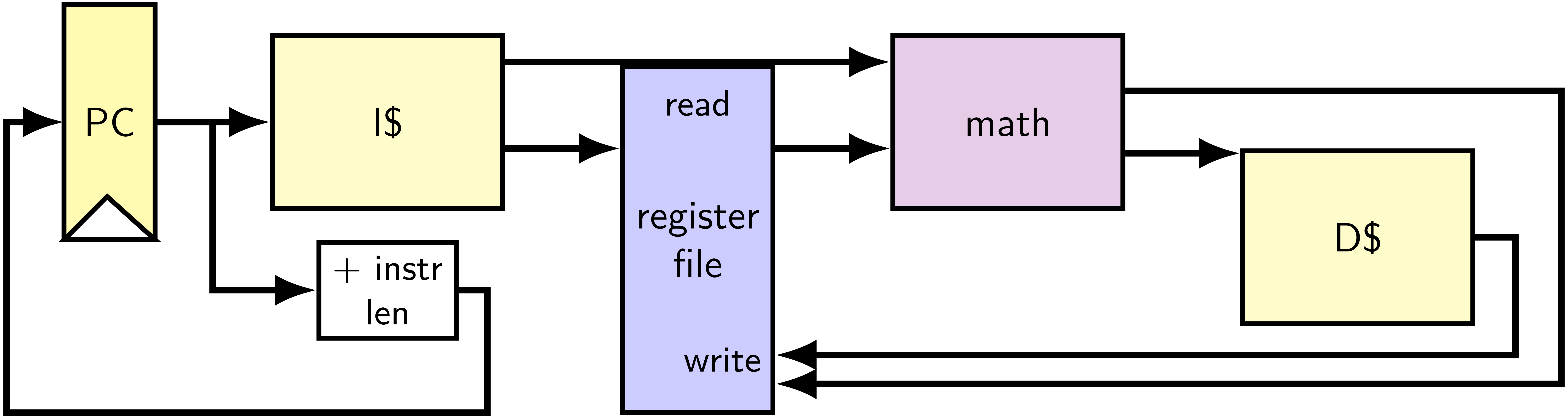
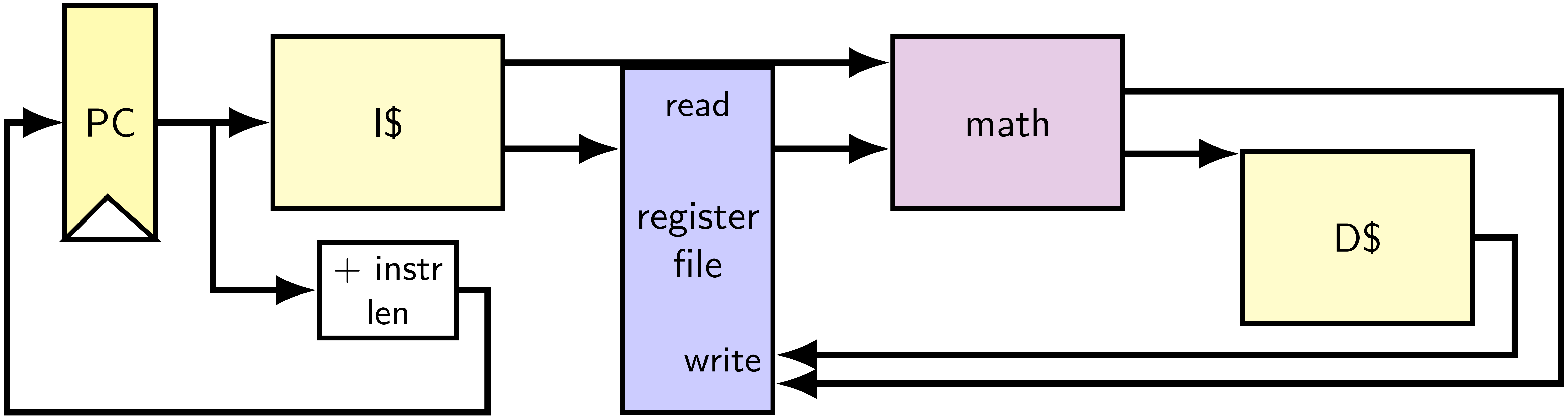
pipeline

simple CPU



running instructions

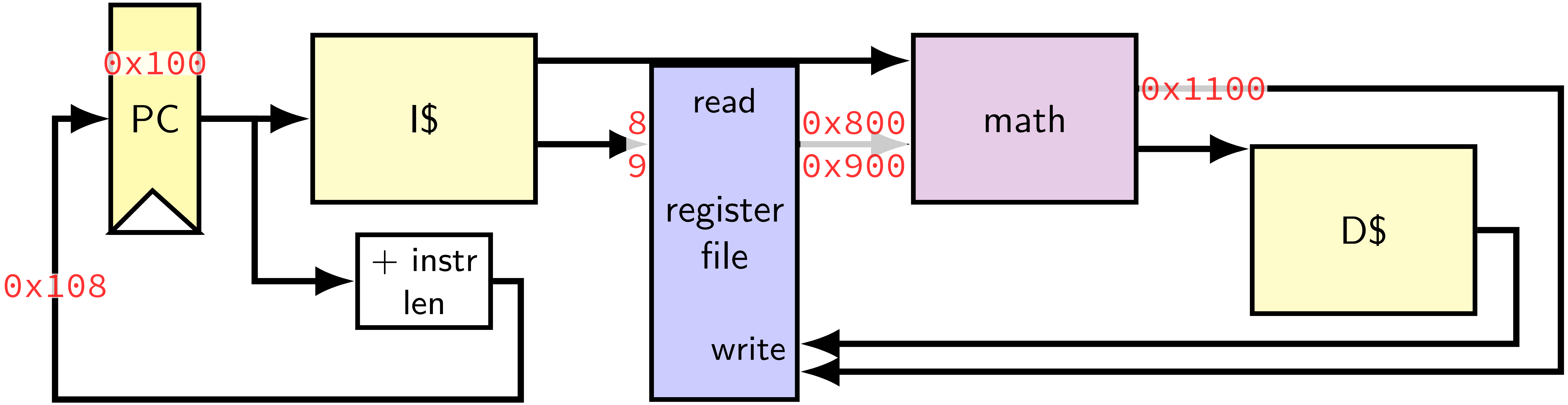
running instructions




```
0x100: addq %r8, %r9
0x108: movq 0x1234(%r10), %r11
```

```
…
%r8: 0x800
%r9: 0x900
%r10: 0x1000
%r11: 0x1100
…
```

running instructions




```
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```
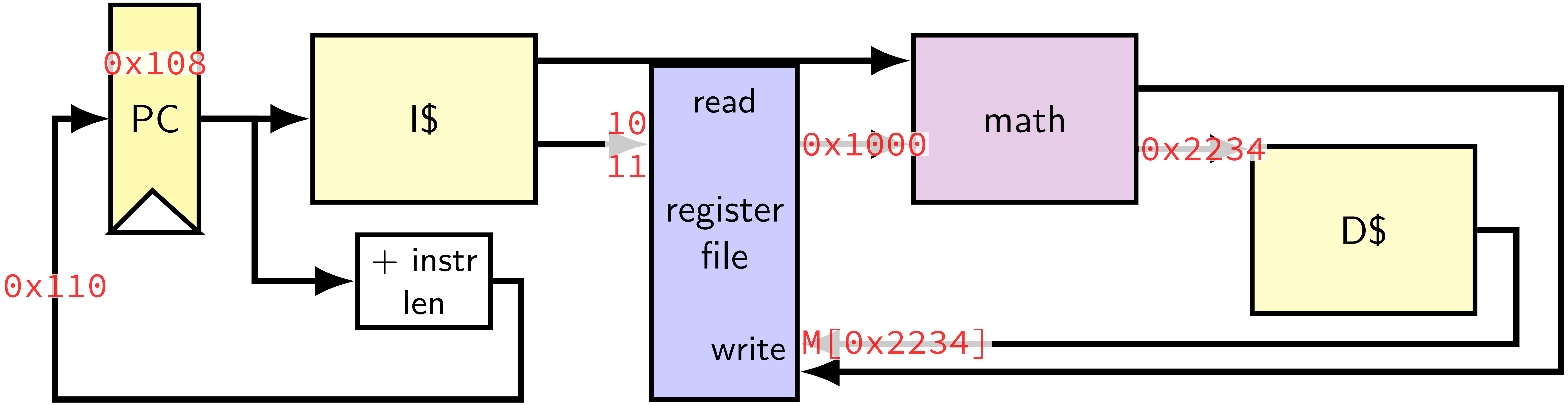

```
…
%r8: 0x800
%r9: 0x1100
%r10: 0x1000
%r11: 0x1100
…
```

running instructions

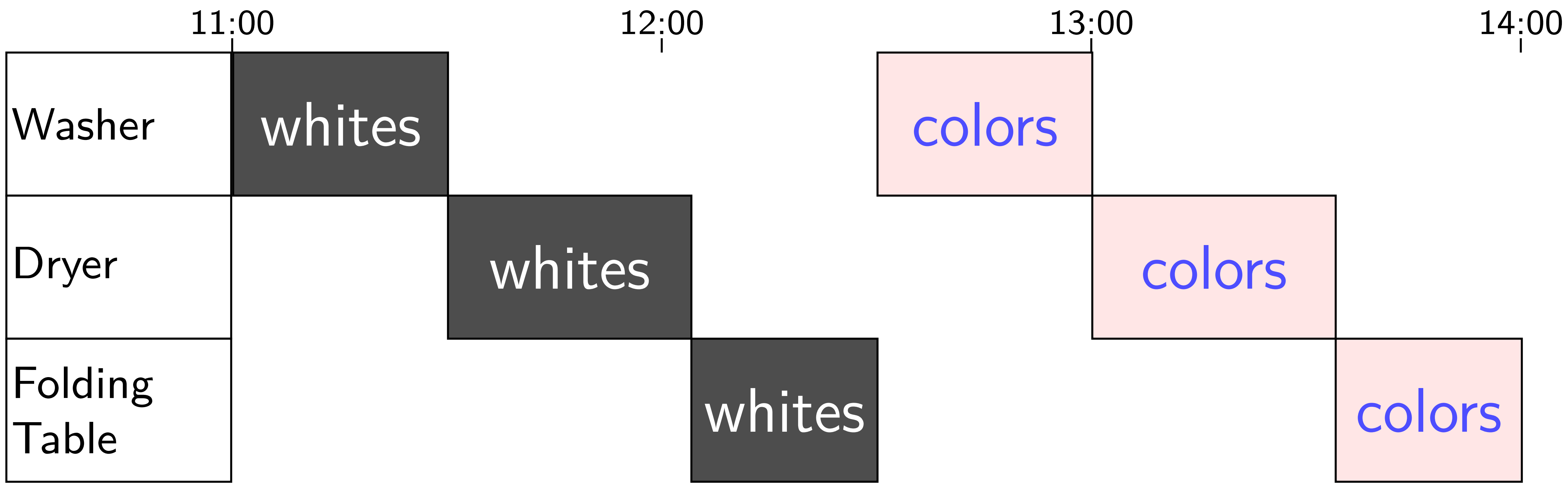



```
0x100: addq %r8, %r9
0x108: movq 0x1234(%r10), %r11
```

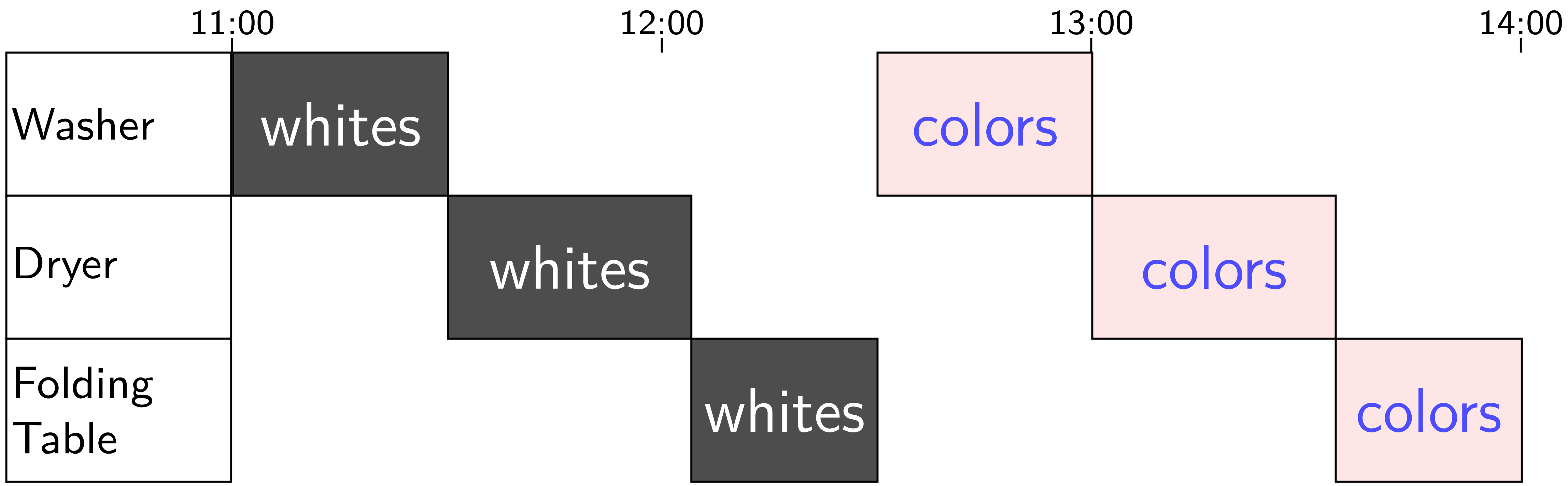
```
…
%r8: 0x800
%r9: 0x1100
%r10: 0x1000
%r11: M[0x2234]
…
```

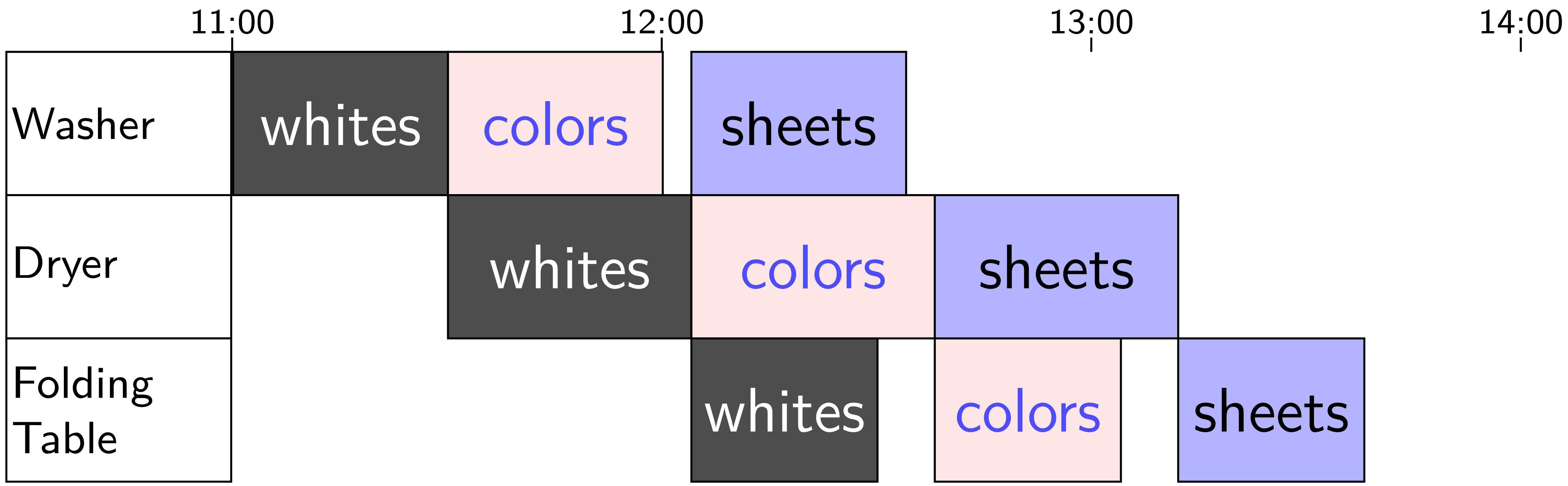
Human pipeline: laundry

Human pipeline: laundry



Human pipeline: laundry





Waste (1)

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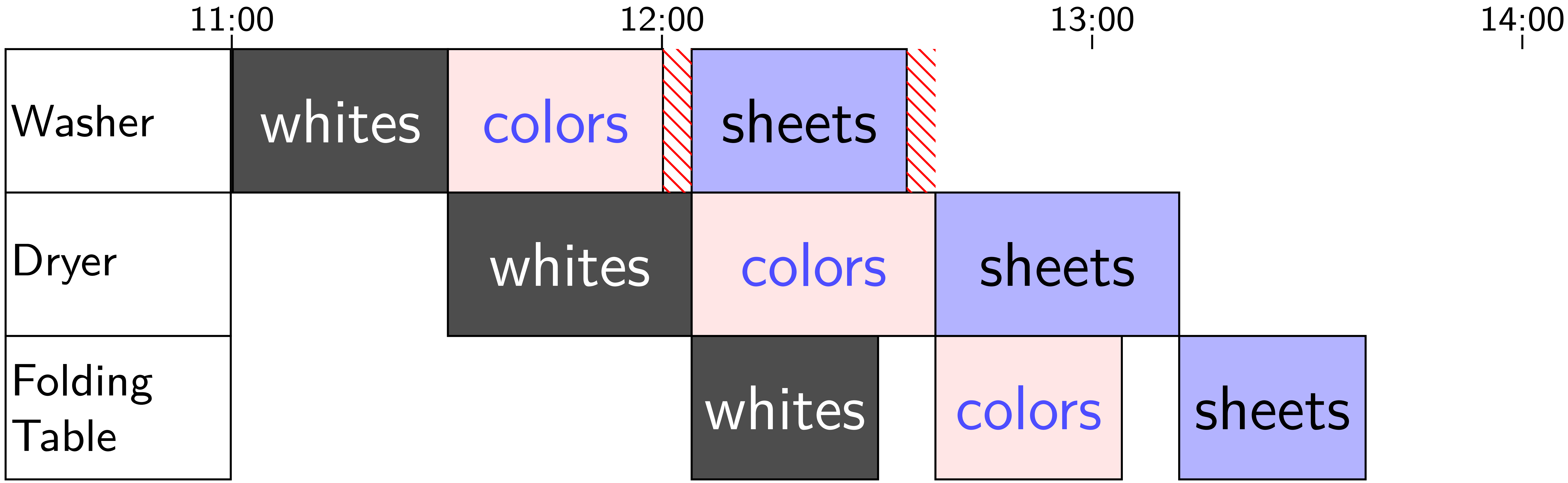
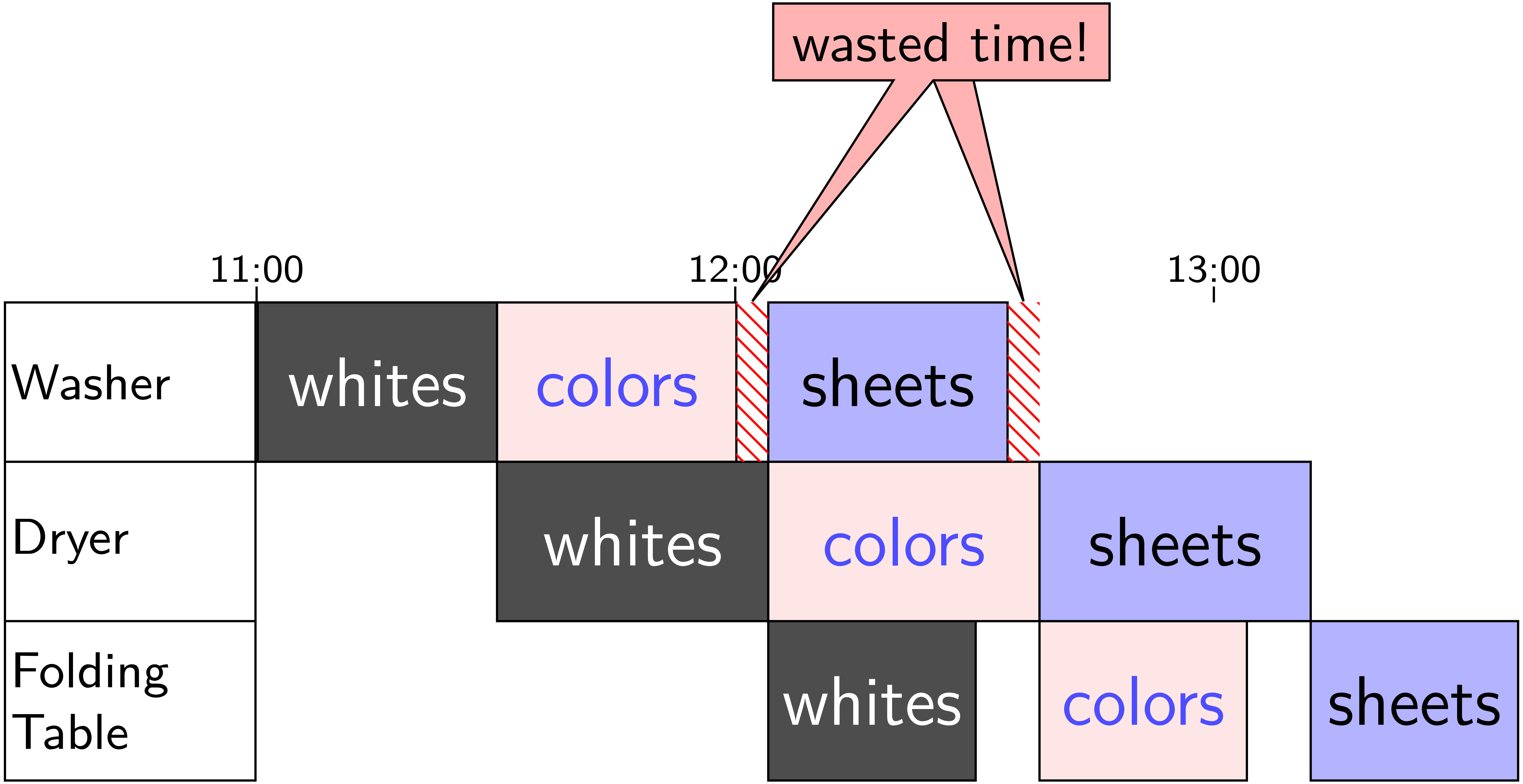
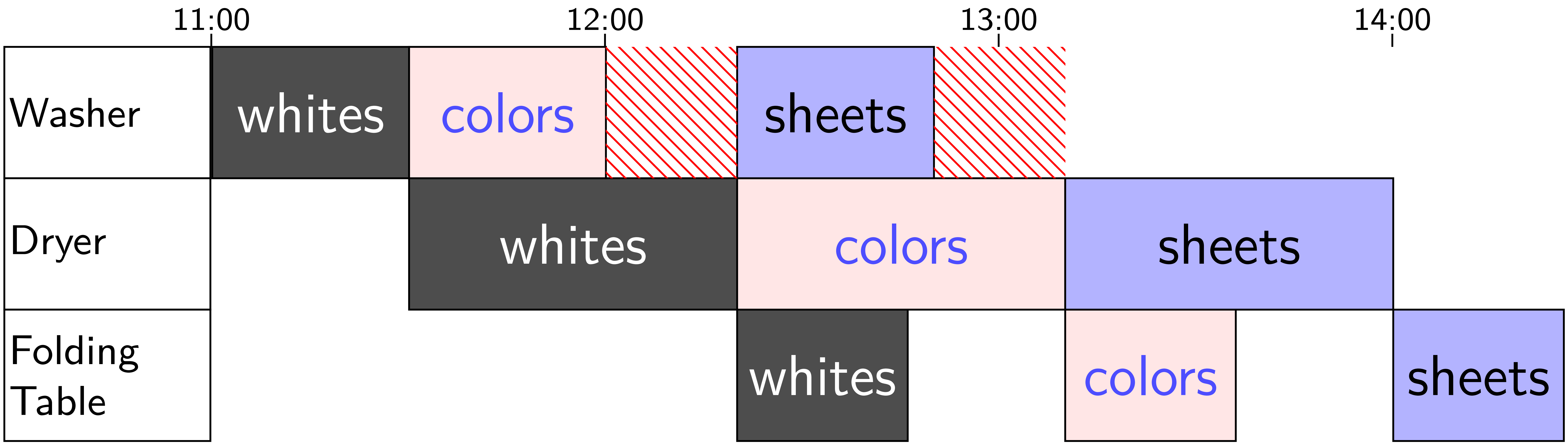

11:00
12:00
13:00
14:00
Washer
whites
colors
sheets
Dryer
whites
colors
sheets
Folding Table
whites
colors
sheets

Waste (1)



wasted time!
11:00
12:00
13:00
14:00
Washer
whites
colors
sheets
Dryer
whites
colors
sheets
Folding Table
whites
colors
sheets

Waste (2)


11:00
12:00
13:00
14:00
Washer
whites
colors
sheets
Dryer
whites
colors
sheets
Folding Table
whites
colors
sheets

Latency — Time for One

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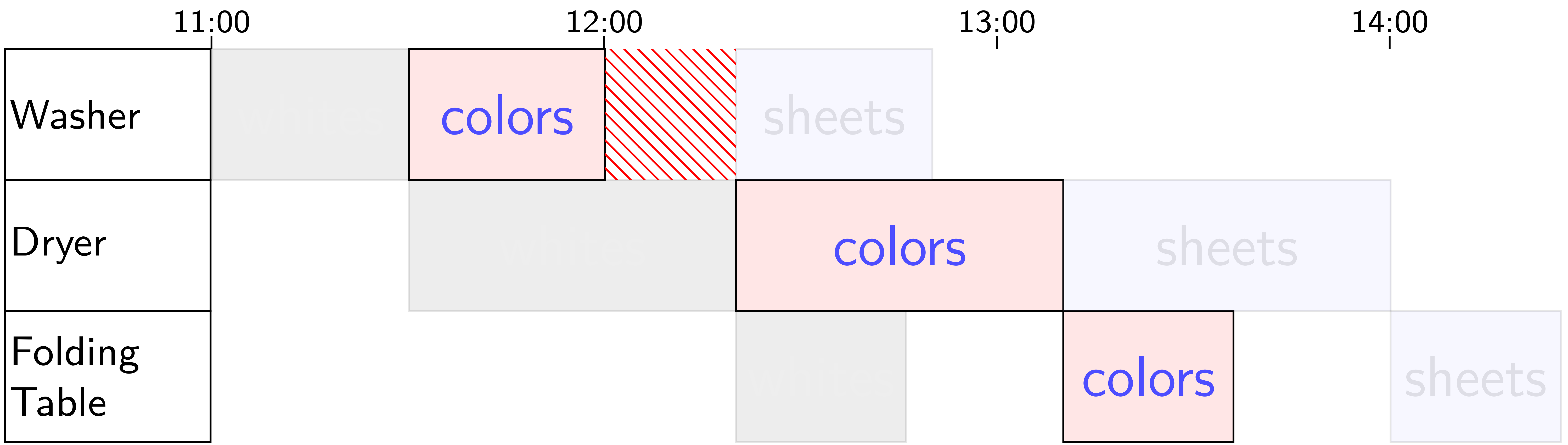
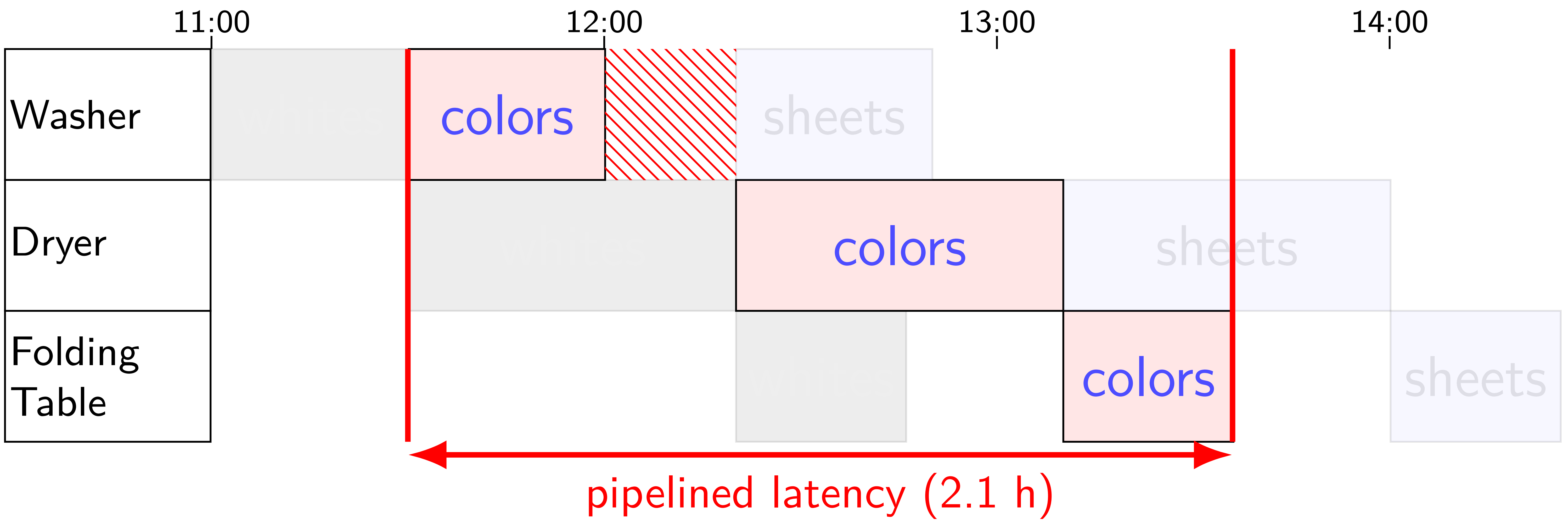
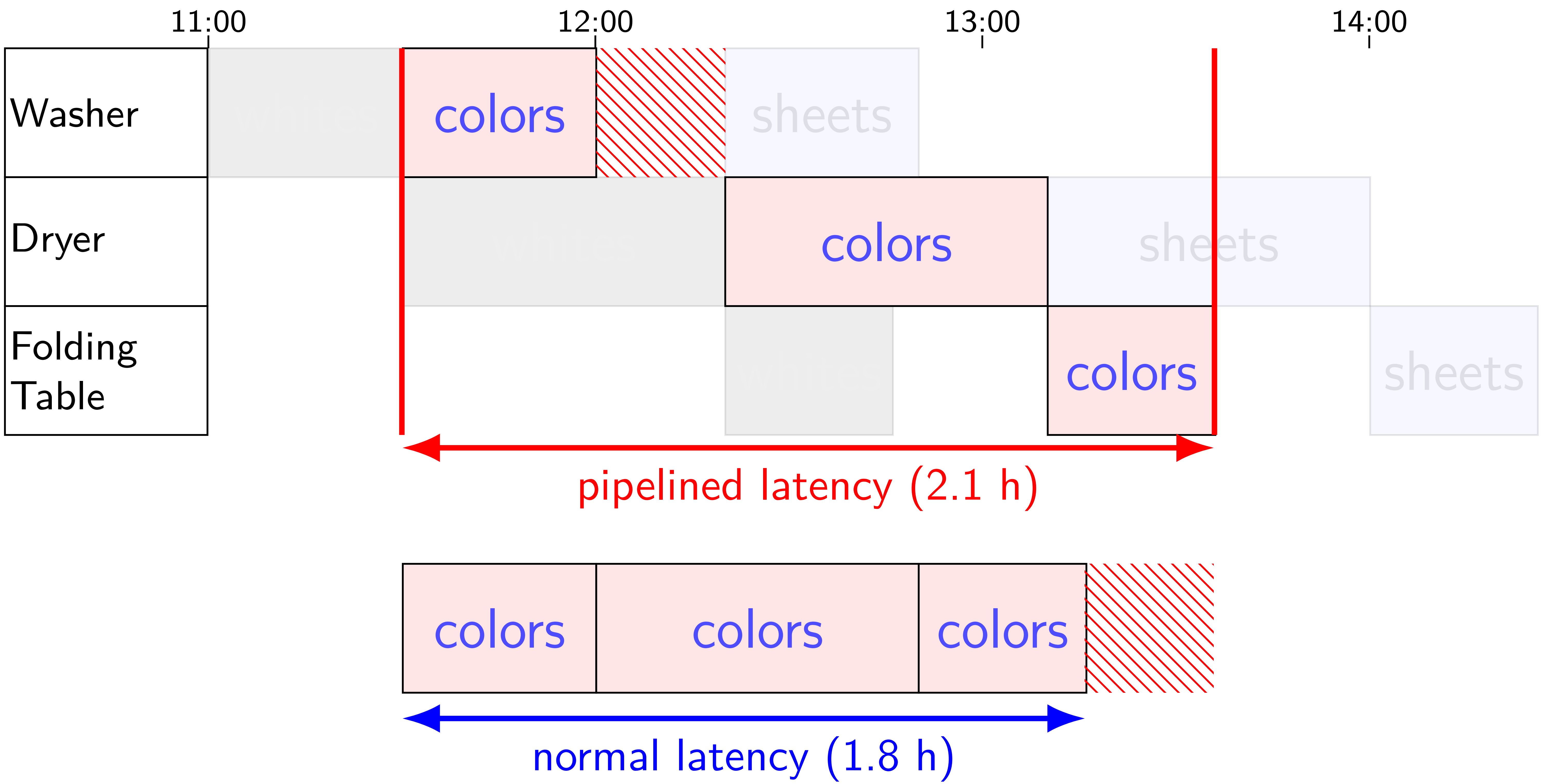


Latency — Time for One



Latency — Time for One



Throughput — Rate of Many

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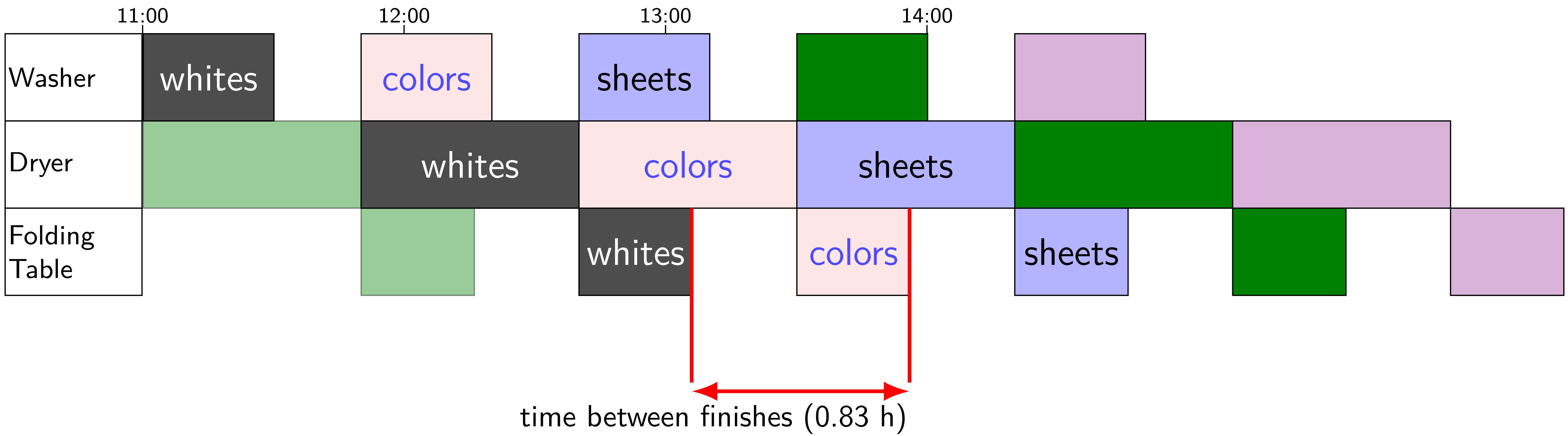
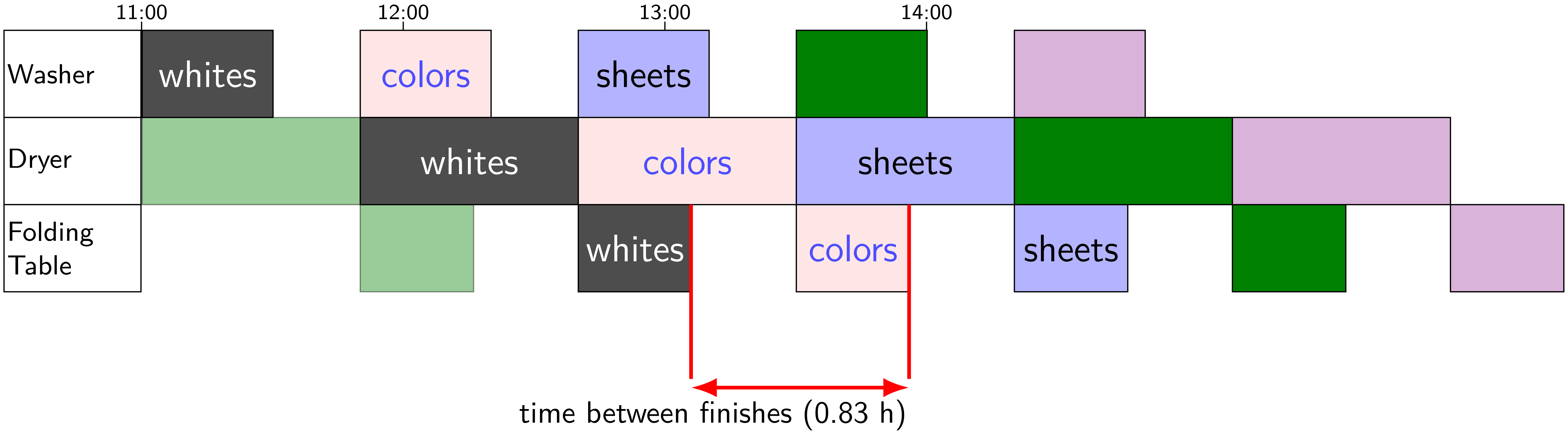
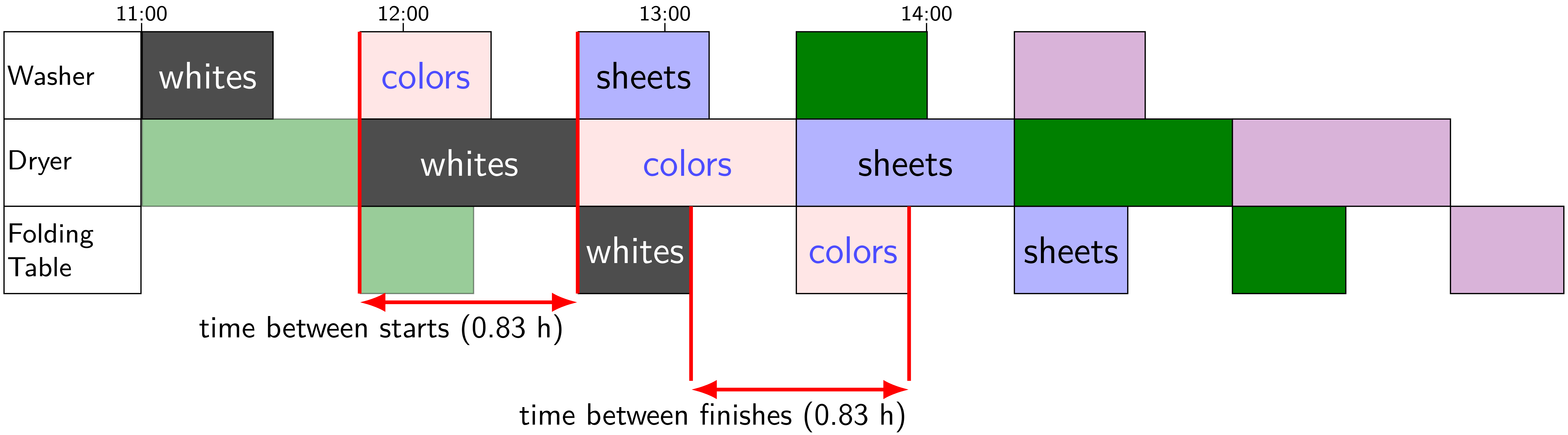


Throughput — Rate of Many



$$\frac{1\ \text{load}}{0.83\text{h}} = 1.2\ \text{loads/h}$$

Throughput — Rate of Many



$$\frac{1 \text{ load}}{0.83\text{h}} = 1.2 \text{ loads/h}$$

adding stages (one way)

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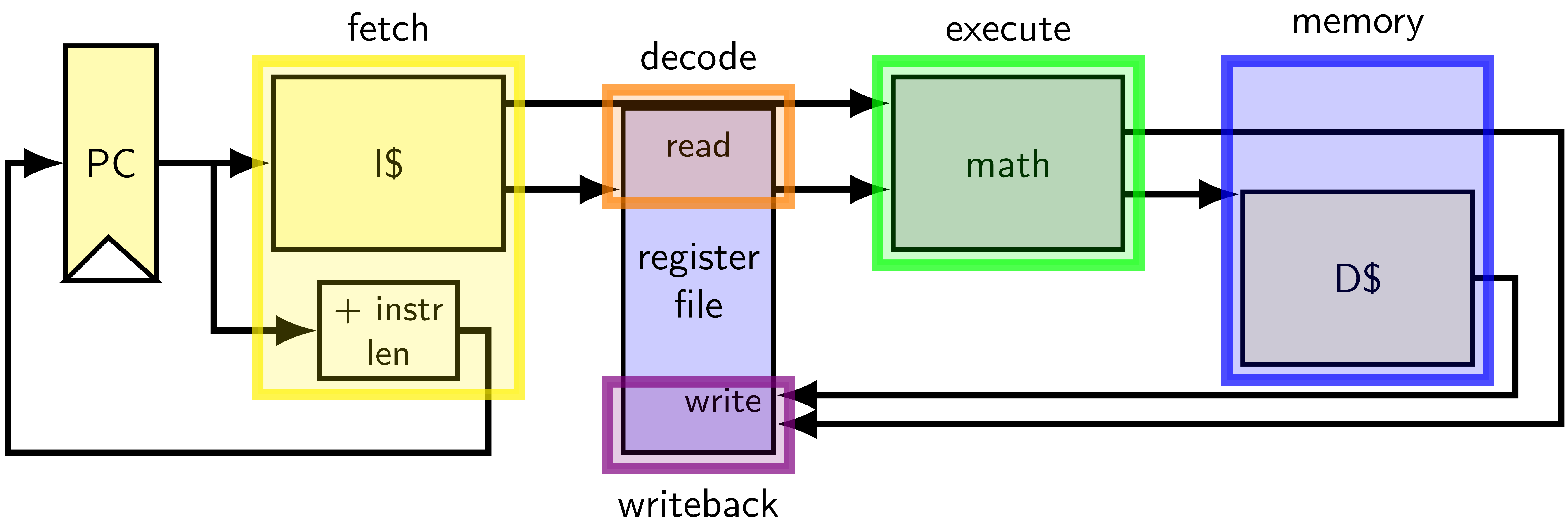
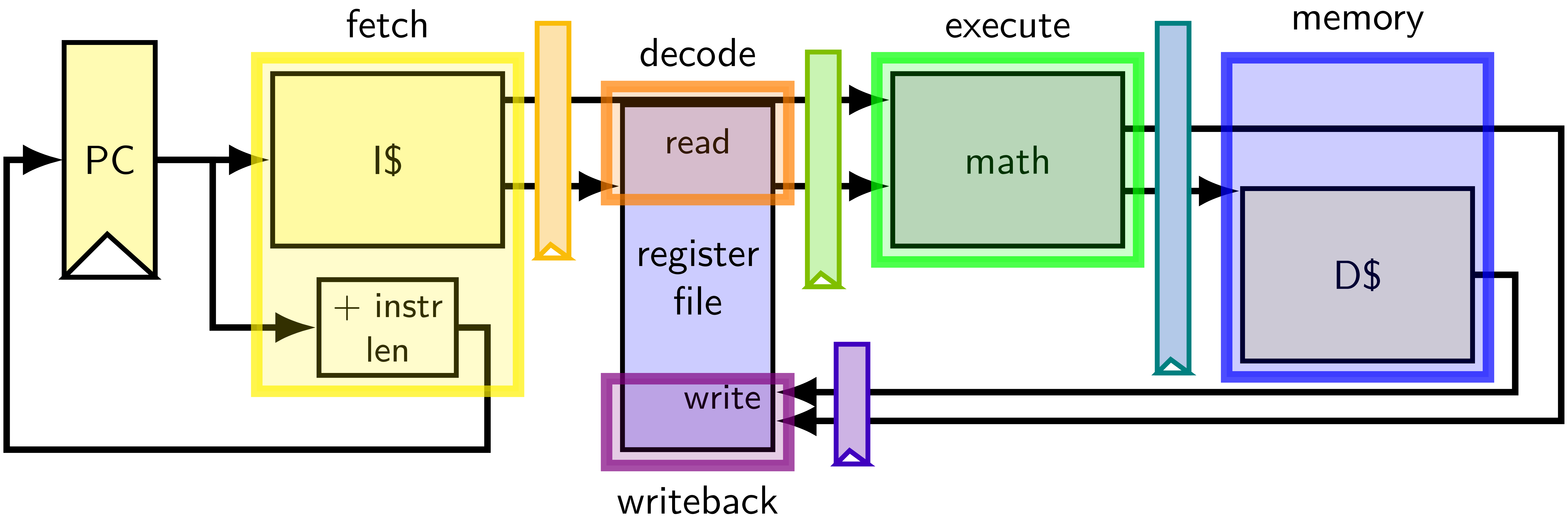


divide running instruction into steps
one way: fetch / decode / execute / memory / writeback

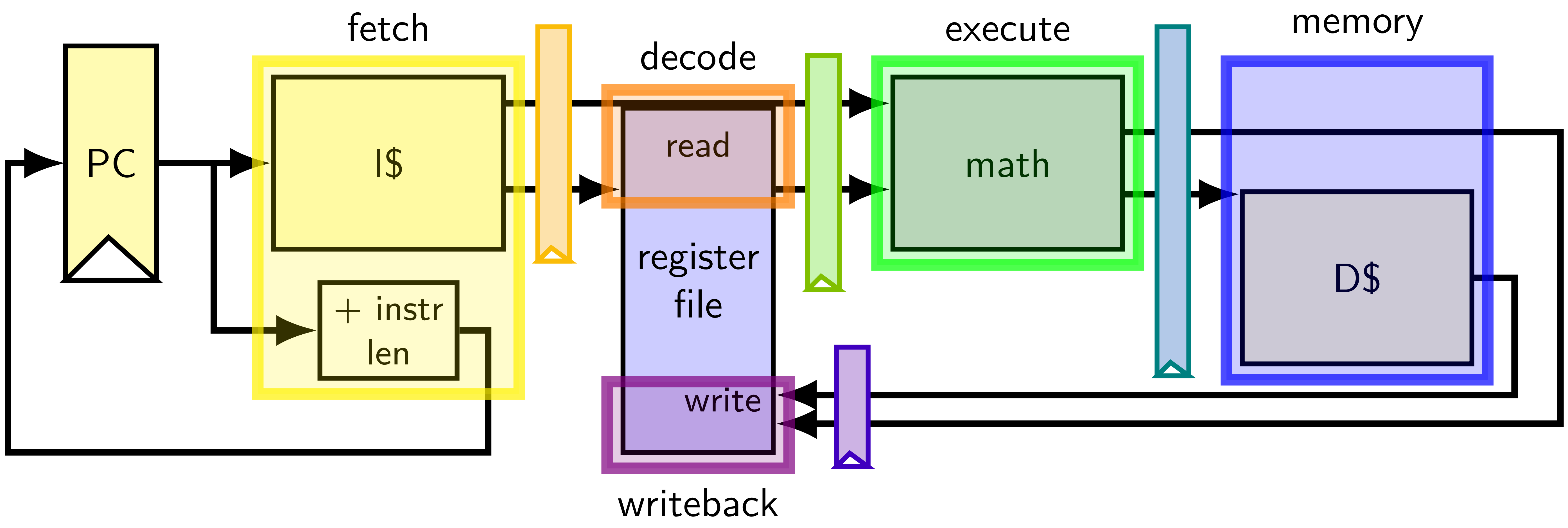
adding stages (one way)



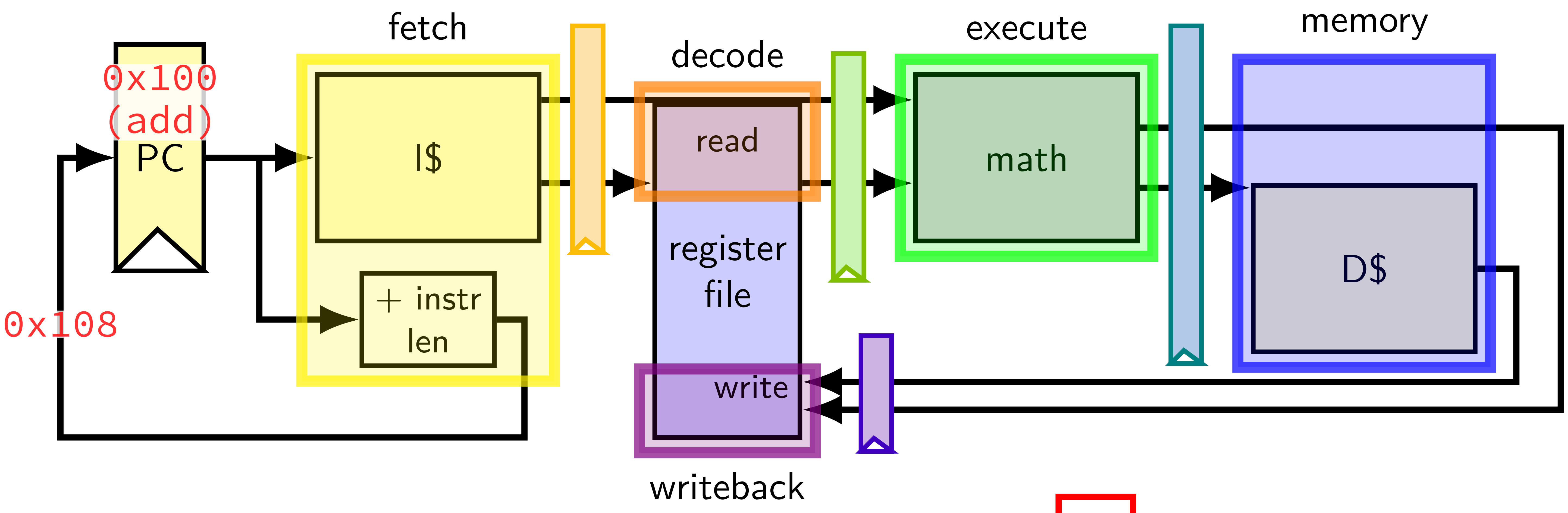
add 'pipeline registers' to hold values from instruction

running some instructions

running some instructions



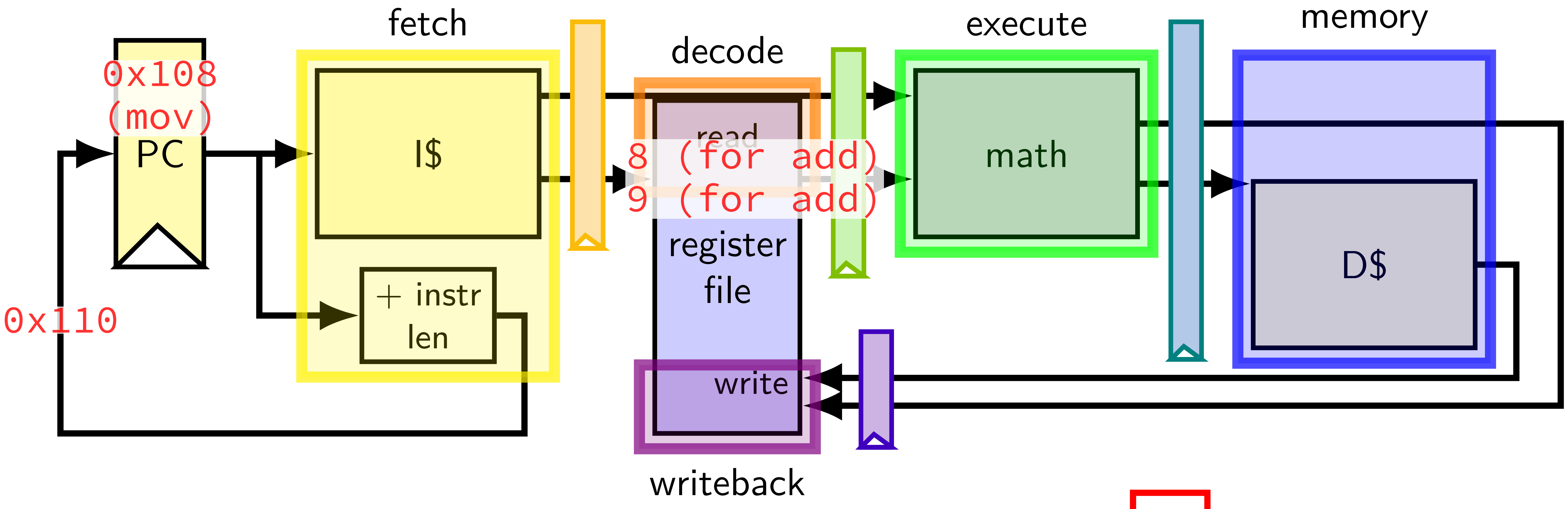
running some instructions




```
0x100: add %r8, %r9
0x108: mov 0x1234(%r10), %r11
0x110: xor %r12, %r13
```

cycle #	0	1	2	3	4	5	6	7	8
	F	D	E	M	W				
		F	D	E	M	W			
			F	D	E	M	W		

running some instructions

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0x100: add %r8, %r9
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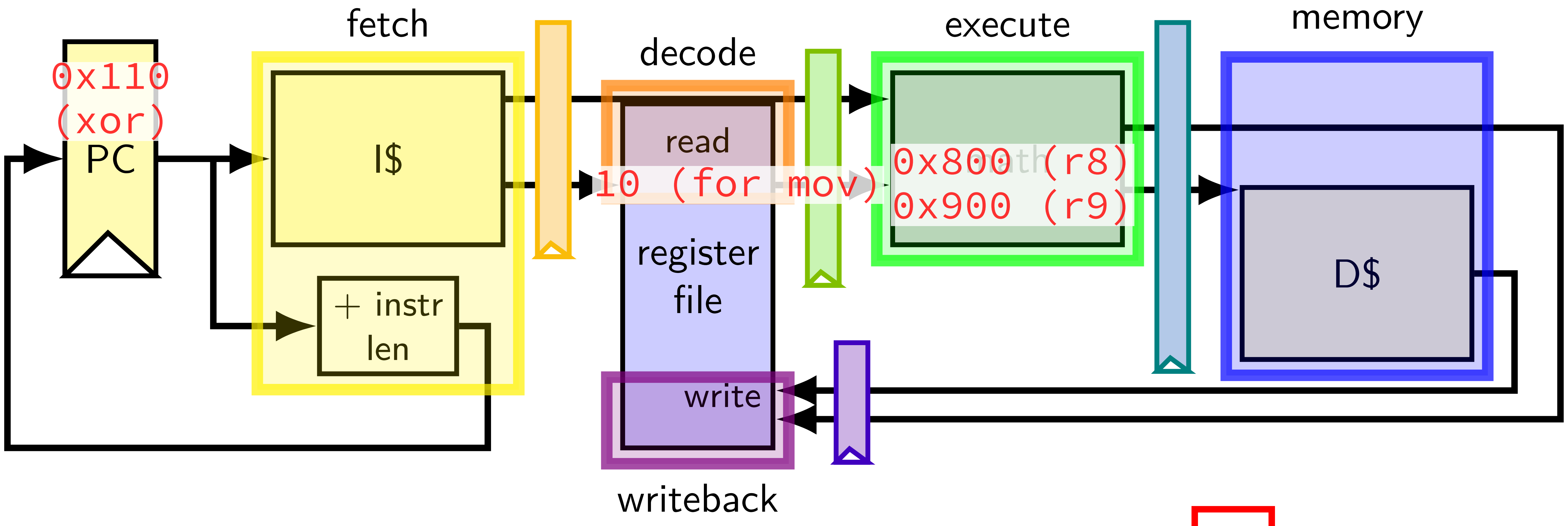
cycle #	0	1	2	3	4	5	6	7	8
	F	D	E	M	W				
		F	D	E	M	W			
			F	D	E	M	W		

running some instructions

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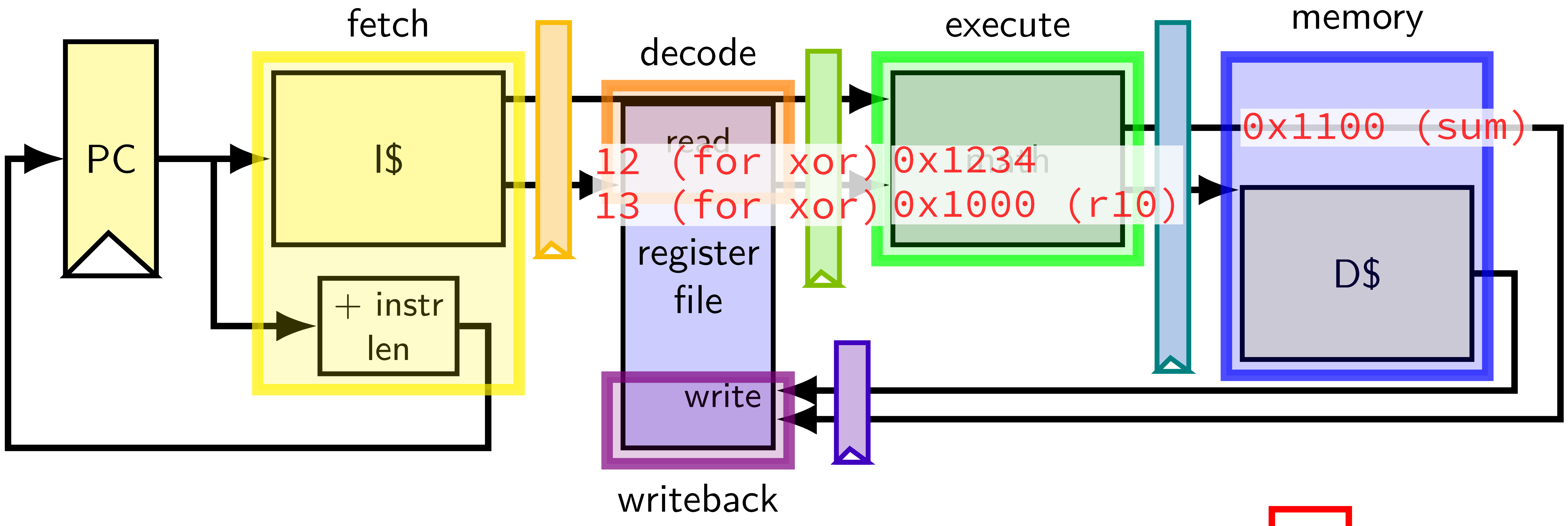
0x100: add %r8, %r9

0x108: mov 0x1234(%r10), %r11

0x110: xor %r12, %r13
```

cycle #	0	1	2	3	4	5	6	7	8
	F	D	E	M	W				
		F	D	E	M	W			
			F	D	E	M	W		

running some instructions

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0x100: add %r8, %r9
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```
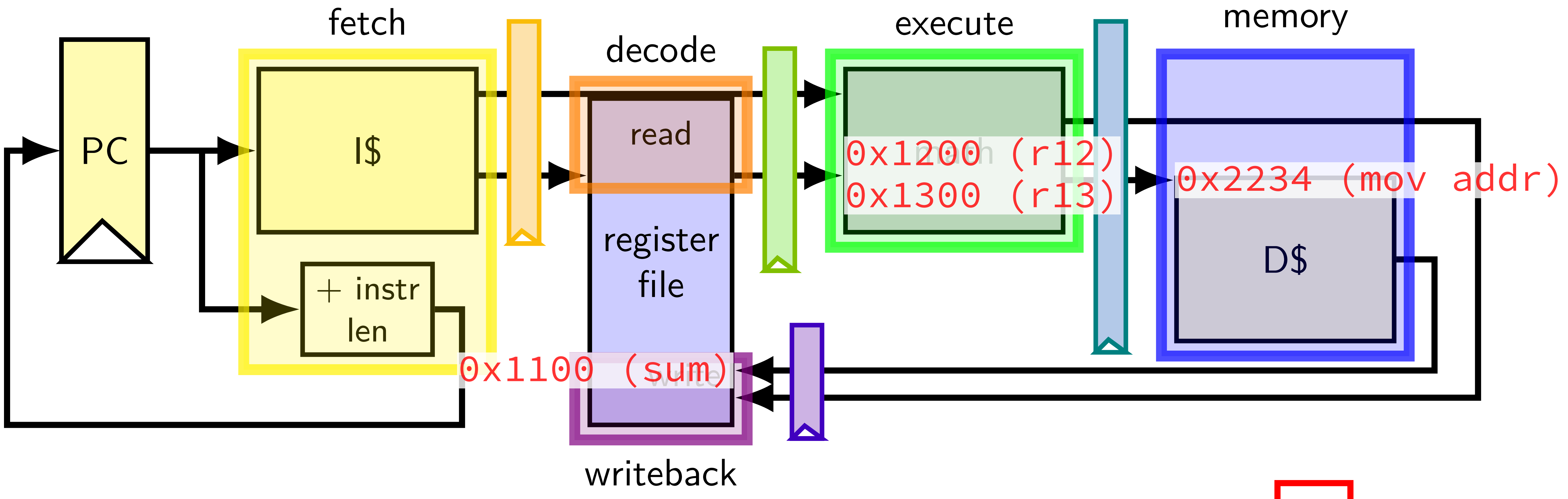

cycle #	0	1	2	3	4	5	6	7	8
	F	D	E	M	W				
		F	D	E	M	W			
			F	D	E	M	W		

running some instructions



cycle #	0	1	2	3	4	5	6	7	8
	F	D	E	M	W				
		F	D	E	M	W			
			F	D	E	M	W		

```
0x100: add %r8, %r9
0x108: mov 0x1234(%r10), %r11
0x110: xor %r12, %r13
```

why registers?

example: fetch/decode

need to store current instruction somewhere … while fetching next one

exercise: throughput/latency (1)

```
0x100: add %r8, %r9
0x108: mov 0x1234(%r10), %r11
0x110: …
```

cycle #	0	1	2	3	4	5	6	7	8
	F	D	E	M	W				
		F	D	E	M	W			
				…					

suppose cycle time is 500 ps

exercise: latency of one instruction?

A. 100 ps B. 500 ps C. 2000 ps D. 2500 ps E. something else

exercise: throughput overall?

A. 1 instr/100 ps B. 1 instr/500 ps C. 1 instr/2000ps D. 1 instr/2500 ps
E. something else

exercise: throughput/latency (2)

```
0x100: add %r8, %r9
0x108: mov 0x1234(%r10), %r11
0x110: …
```

cycle #	0	1	2	3	4	5	6	7	8
	F	D	E	M	W				
		F	D	E	M	W			
				…					

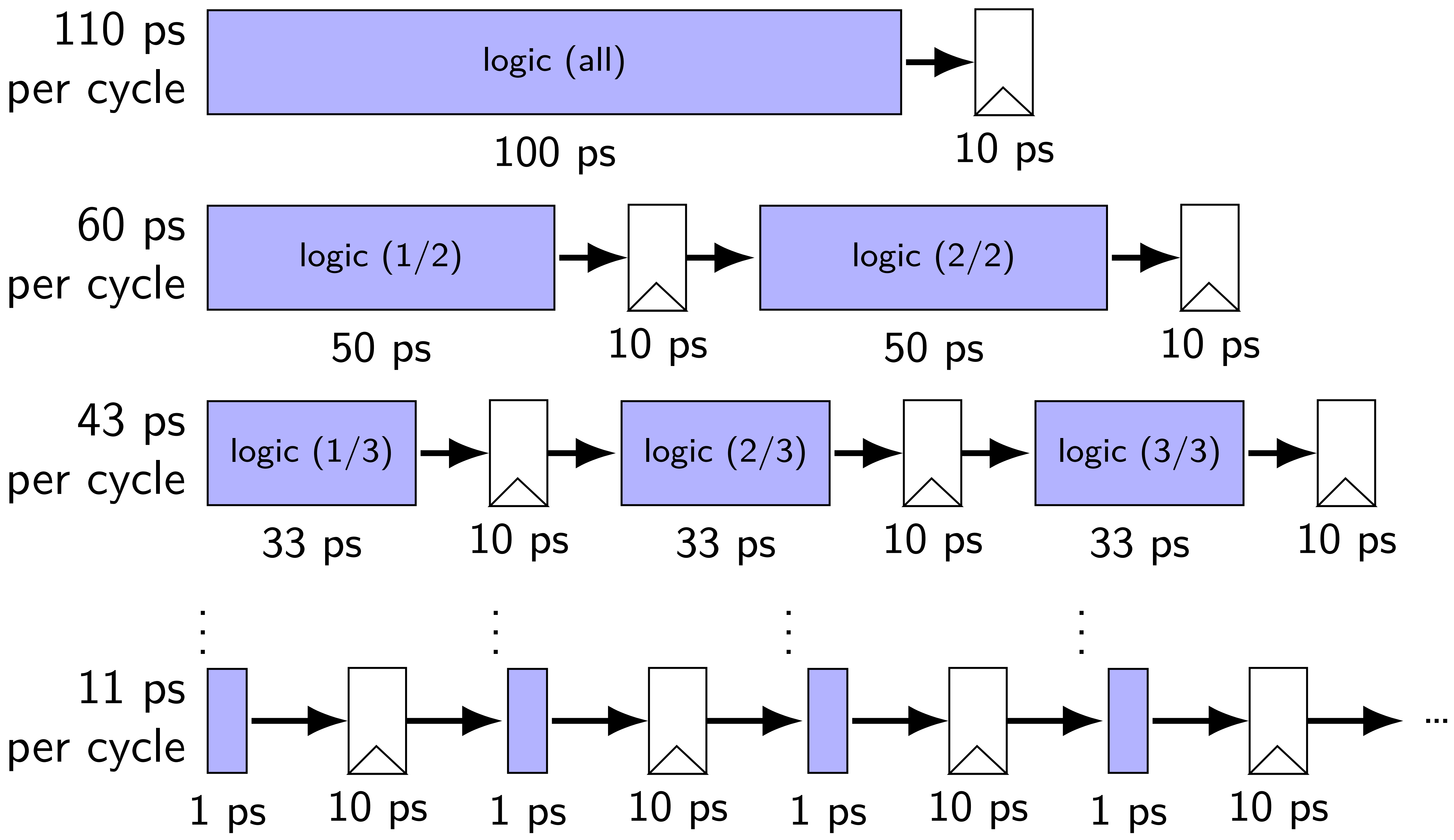
```
0x100: add %r8, %r9
0x108: mov 0x1234(%r10), %r11
0x110: …
```

cycle #	0	1	2	3	4	5	6	7	8	
	F1	F2	D1	D2	E1	E2	M1	M2	W1	W2	
		F1	F2	D1	D2	E1	E2	M1	M2	W1	W2
				…							

double number of pipeline stages (to 10) + decrease cycle time from 500 ps to 250 ps — throughput?

A. 1 instr/100 ps B. 1 instr/250 ps C. 1 instr/1000ps D. 1 instr/5000 ps
E. something else

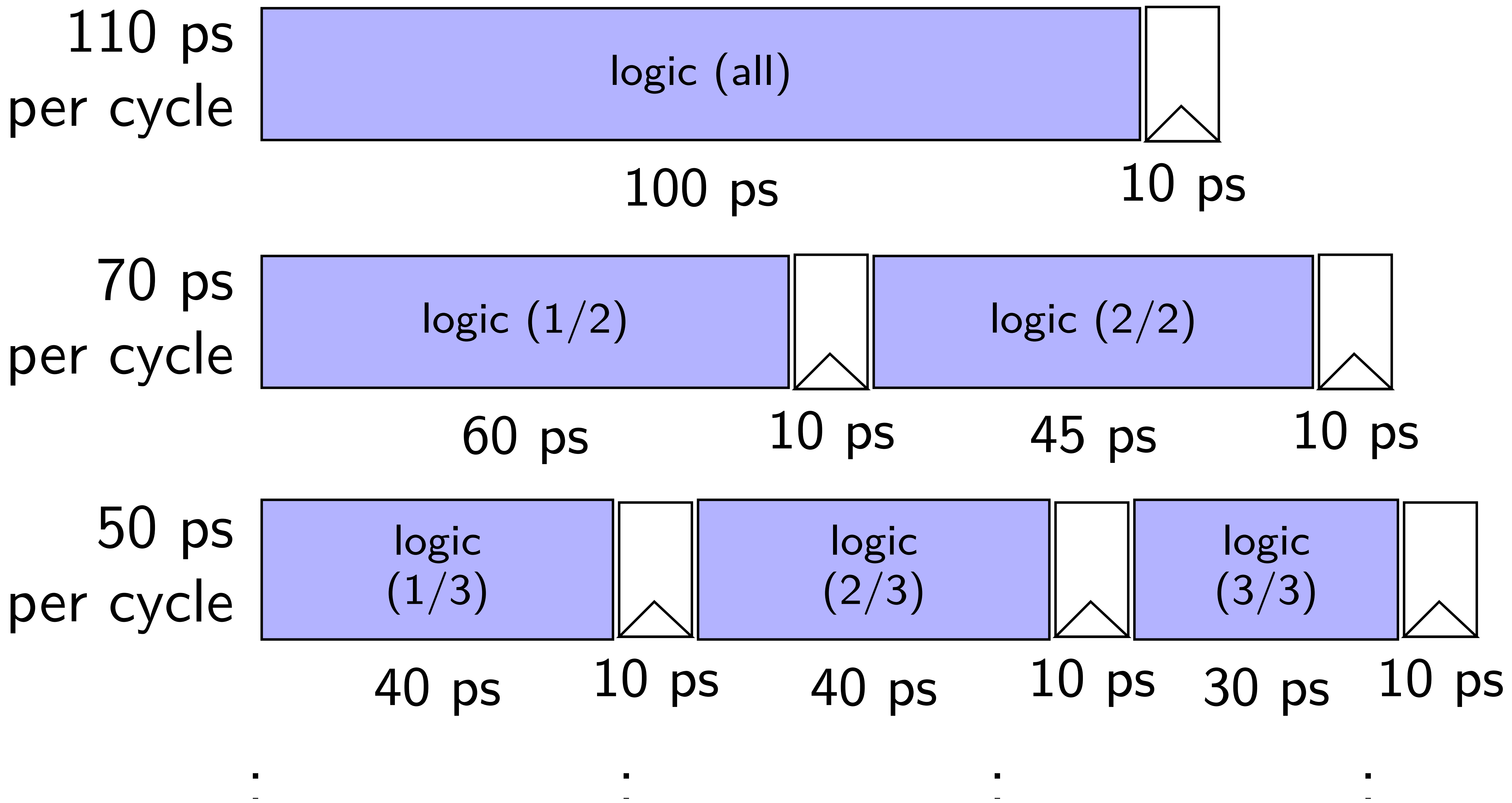
diminishing returns: register delays



diminishing returns: uneven split

Can we split up some logic (e.g. adder) arbitrarily?

Probably not...



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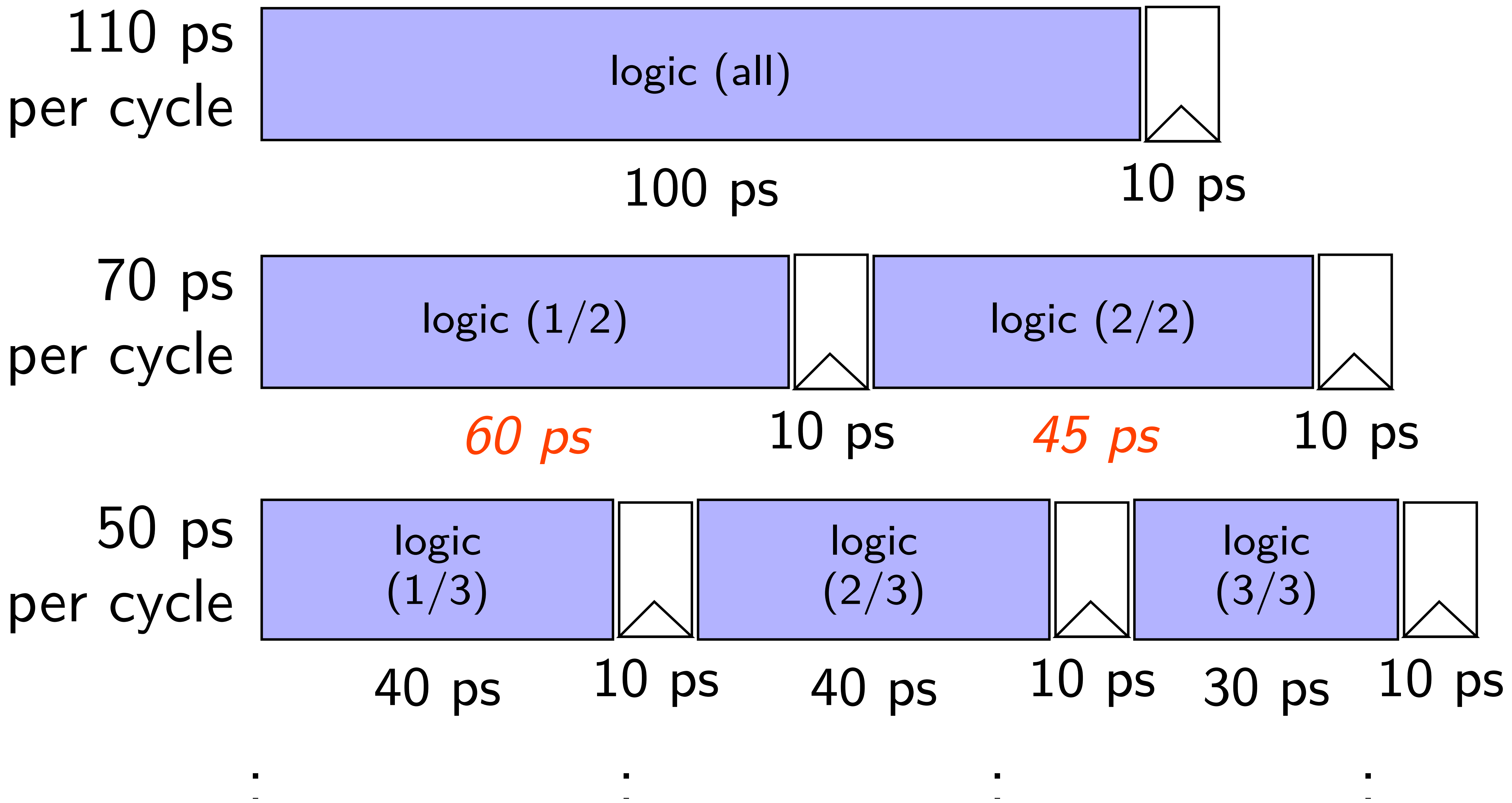

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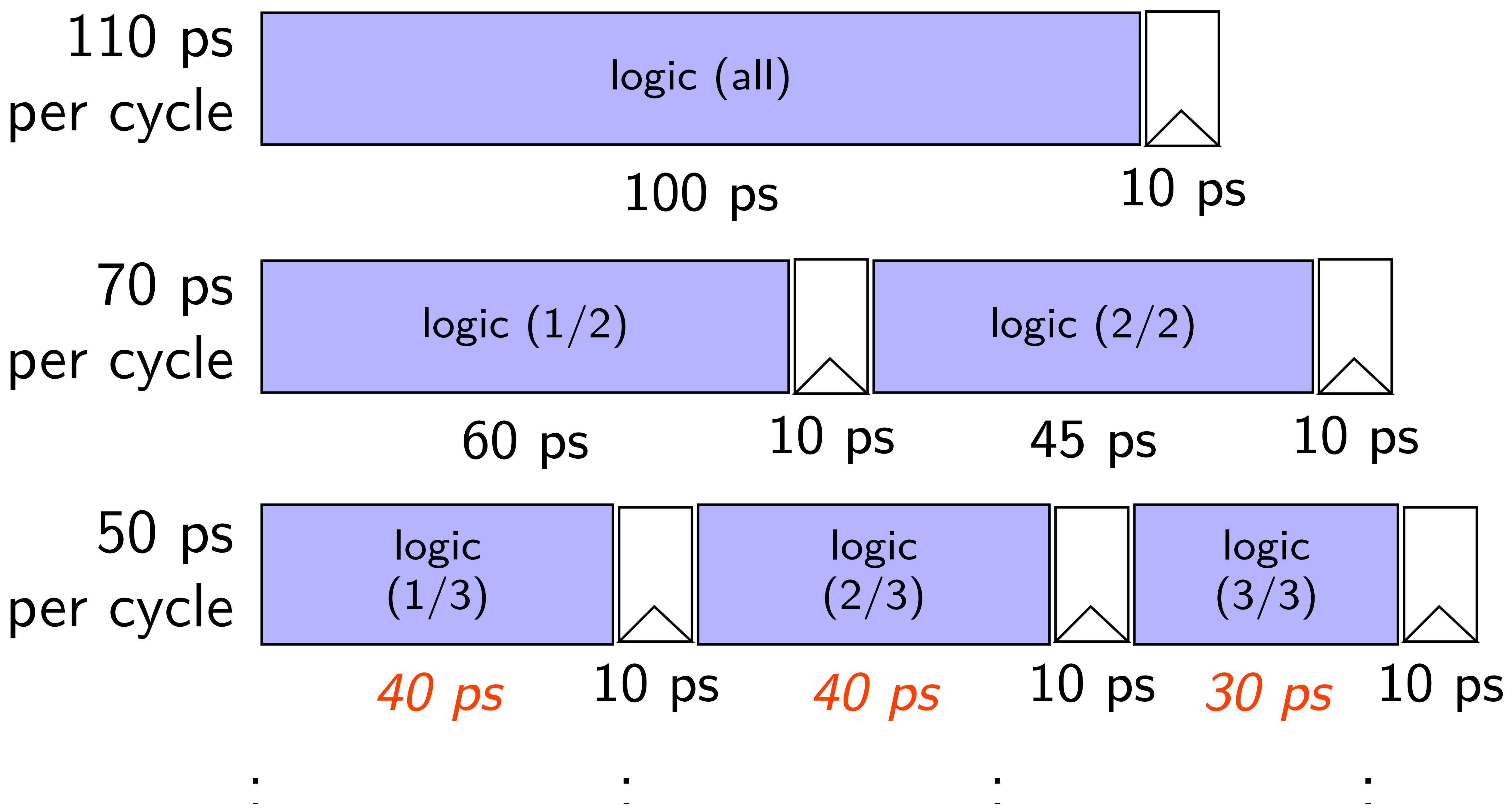
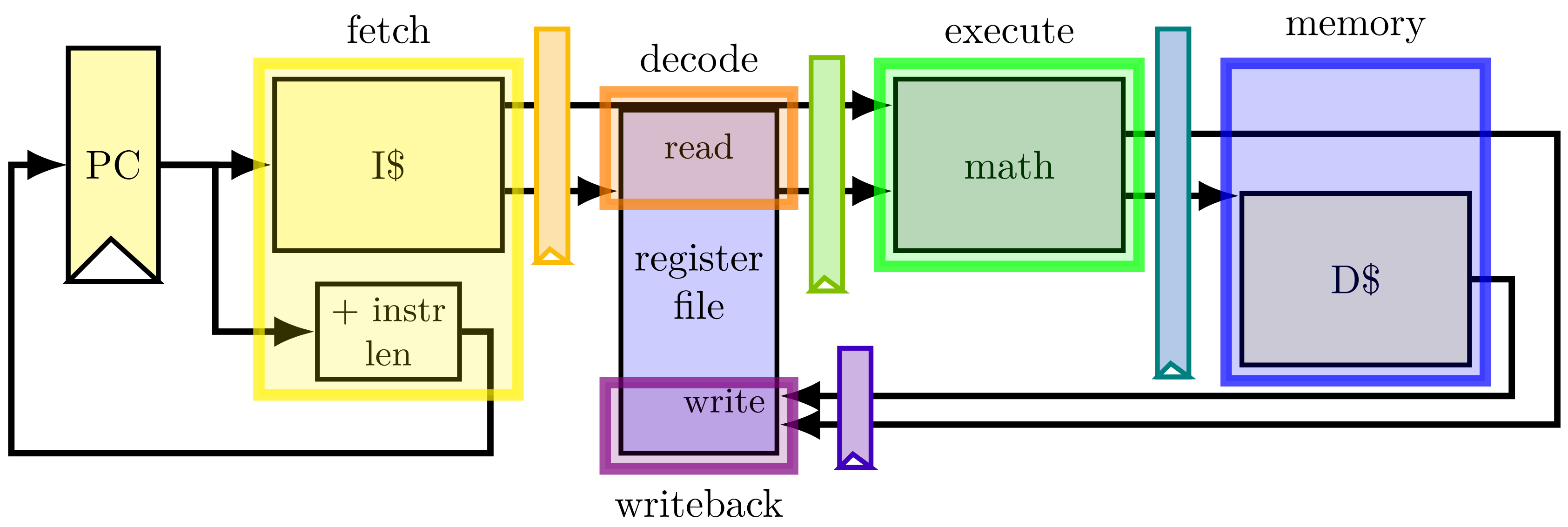


a data hazard

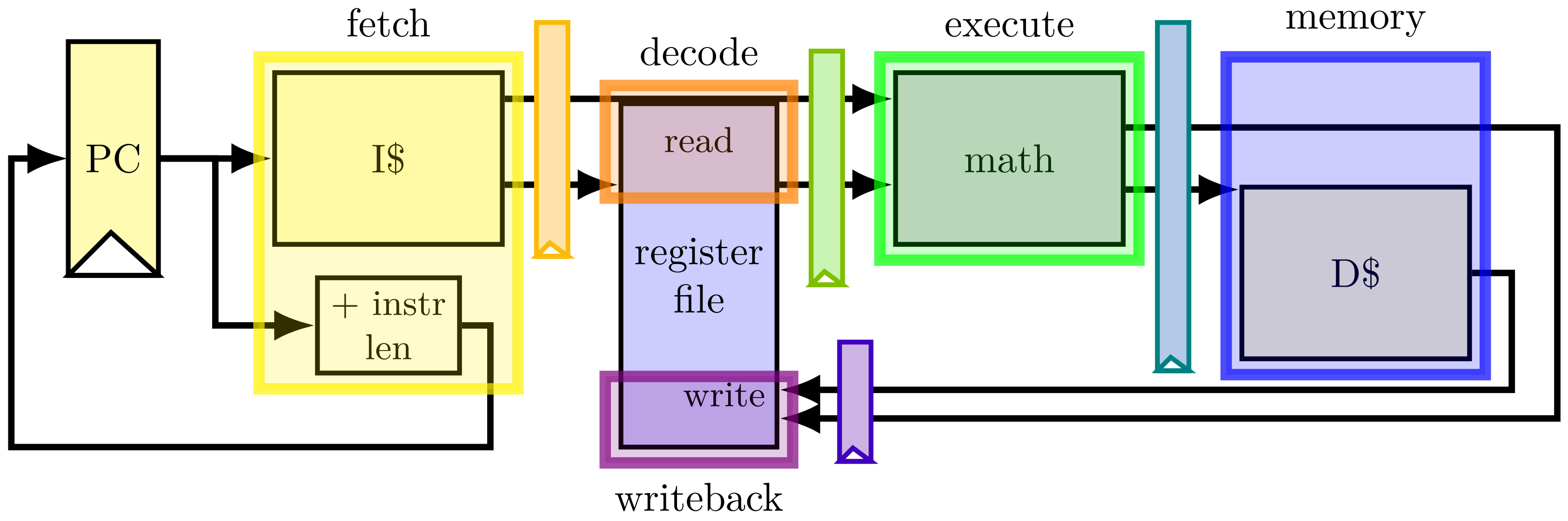
```
addq %r8, %r9    // R8 + R9 -> R9
addq %r9, %r8    // R9 + R8 -> R9
addq ...
addq ...
```



	fetch	fetch/decode		decode/execute			execute/memory		memory/writeback	
cycle	PC	rA	rB	R[rB]	R[rB]	rB	sum	rB	sum	rB
0	`0x0`									
1	`0x2`	8	9							
2		9	8	800	900	9				
3				900	800	8	1700	9		
4							1700	8	1700	9
5									1700	8

a data hazard

```
addq %r8, %r9    // R8 + R9 -> R9
addq %r9, %r8    // R9 + R8 -> R9
addq ...
addq ...
```



	fetch	fetch/decode		decode/execute			execute/memory		memory/writeback	
cycle	PC	rA	rB	R[rB]	R[rB]	rB	sum	rB	sum	rB
0	0x0									
1	0x2	8	9							
2		9	8	800	900	9				
3				900	800	8	1700	9		
4							1700	8	1700	9
5									1700	8

should be 1700

data hazard

```
addq %r8, %r9  // (1)
addq %r9, %r8  // (2)
```

step#	pipeline implementation	ISA specification
1	read r8, r9 for (1)	read r8, r9 for (1)
2	read r9, r8 for (2)	write r9 for (1)
3	write r9 for (1)	read r9, r8 for (2)
4	write r8 for (2)	write r8 ror (2)

pipeline reads *older value*…

instead of value ISA says was just written

data hazard compiler solution

```
addq %r8, %r9
nop
nop
addq %r9, %r8
```

one solution: *change the ISA*

- all `addq`s take effect *three instructions later*
- (assuming can read register value while it is being written back)

make it *compiler's job*

problem: recompile everytime processor changes?

data hazard compiler solution

```
addq %r8, %r9
nop
nop
addq %r9, %r8
```

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make it *compiler's job*

problem: recompile everytime processor changes?

stalling/nop pipeline diagram (1)

stalling/nop pipeline diagram (1)

	cycle # 0	1	2	3	4	5	6	7	8
`add %r8, %r9`	F	D	E	M	W				
`nop`		F	D	E	M	W			
`nop`			F	D	E	M	W		
`addq %r9, %r8`				F	D	E	M	W	

stalling/nop pipeline diagram (1)

	cycle # 0	1	2	3	4	5	6	7	8
`add %r8, %r9`	F	D	E	M	W				
`nop`		F	D	E	M	W			
`nop`			F	D	E	M	W		
`addq %r9, %r8`				F	D	E	M	W	

assumption:
if writing register value
register file will return that value for reads

not actually way register file worked in single-cycle CPU
(e.g. can read old %r9 while writing new %r9)

stalling/nop pipeline diagram (2)

stalling/nop pipeline diagram (2)

cycle #	0	1	2	3	4	5	6	7	8
`add %r8, %r9`	F	D	E	M	W				
`nop`		F	D	E	M	W			
`nop`			F	D	E	M	W		
`nop`				F	D	E	M	W	
`addq %r9, %r8`					F	D	E	M	W

stalling/nop pipeline diagram (2)

	cycle # 0	1	2	3	4	5	6	7	8
`add %r8, %r9`	F	D	E	M	W				
`nop`		F	D	E	M	W			
`nop`			F	D	E	M	W		
`nop`				F	D	E	M	W	
`addq %r9, %r8`					F	D	E	M	W

if we didn't modify the register file, we'd need an extra cycle

data hazard hardware solution

```
addq %r8, %r9
// hardware inserts: nop
// hardware inserts: nop
addq %r9, %r8
```

how about hardware add nops?

called *stalling*

extra logic:

- sometimes don't change PC
- sometimes put do-nothing values in pipeline registers

control hazard

```
0x00: cmpq %r8, %r9
0x08: je   0xFFFF
0x10: addq %r10, %r11
```

	fetch	fetch→decode		decode→execute		execute→writeb	execute→writeback		…
cycle	PC	rA	rB	R[rA]	R[rB]	result	…	…	…
0	0x0								
1	0x8	8	9						
2	???	---	---	800	900				
3	???	---	---	---	---	less than			

control hazard

```
0x00: cmpq %r8, %r9
0x08: je   0xFFFF
0x10: addq %r10, %r11
```

	fetch	fetch→decode		decode→execute		execute→writeb	execute→writeback		…
cycle	PC	rA	rB	R[rA]	R[rB]	result	…	…	…
0	0x0								
1	0x8	8	9						
2	???	---	---	800	900				
3	???	---	---	---	---	less than			

`0xFFFF` if R[8] = R[9]; `0x10` otherwise

jXX: stalling?

```
cmpq %r8, %r9
jne LABEL // not taken
xorq %r10, %r11
movq %r11, 0(%r12)
...
```

jXX: stalling?

```
cmpq %r8, %r9
jne LABEL // not taken
xorq %r10, %r11
movq %r11, 0(%r12)
...
```

cycle #	0	1	2	3	4	5	6	7	8	
cmpq %r8, %r9	F	D	E	M	W					
jne LABEL		F	D	E	M	W				
(do nothing)			F	D	E	M	W			
(do nothing)				F	D	E	M	W		
xorq %r10, %r11					F	D	E	M	W	
movq %r11, 0(%r12)						F	D	E	M	W
…										

jXX: stalling?

```
cmpq %r8, %r9
jne LABEL // not taken
xorq %r10, %r11
movq %r11, 0(%r12)
...
```

	cycle # 0	1	2	3	4	5	6	7	8
cmpq %r8, %r9	F	D	E	M	W				
jne LABEL		F	D	E	M	W			
(do nothing)			F	D	E	M	W		
(do nothing)				F	D	E	M	W	
xorq %r10, %r11					F	D	E	M	W
movq %r11, 0(%r12)						F	D	E	M W
…									

compare sets flags

jXX: stalling?

```
cmpq %r8, %r9
jne LABEL // not taken
xorq %r10, %r11
movq %r11, 0(%r12)
...
```

compute if jump goes to LABEL

cycle #	0	1	2	3	4	5	6	7	8	
cmpq %r8, %r9	F	D	E	M	W					
jne LABEL		F	D	$\boxed{E}$	M	W				
(do nothing)			F	D	E	M	W			
(do nothing)				F	D	E	M	W		
xorq %r10, %r11					F	D	E	M	W	
movq %r11, 0(%r12)						F	D	E	M	W
…										

jXX: stalling?

```
cmpq %r8, %r9
jne LABEL // not taken
xorq %r10, %r11
movq %r11, 0(%r12)
...
```

---

	cycle # 0	1	2	3	4	5	6	7	8	
cmpq %r8, %r9	F	D	E	M	W					
jne LABEL		F	D	E	M	W				
(do nothing)			F	D	E	M	W			
(do nothing)				F	D	E	M	W		
xorq %r10, %r11				use computed result	F	D	E	M	W	
movq %r11, 0(%r12)						F	D	E	M	W
…										

making guesses

```
        cmpq %r8, %r9
        jne LABEL
        xorq %r10, %r11
        movq %r11, 0(%r12)
        ...

LABEL:  addq %r8, %r9
        imul %r13, %r14
        ...
```

speculate (guess): *jne* won't go to LABEL

right: 2 cycles faster!; wrong: undo guess before too late

jXX: speculating right (1)

```
        cmpq %r8, %r9
        jne LABEL
        xorq %r10, %r11
        movq %r11, 0(%r12)
        ...

LABEL:  addq %r8, %r9
        imul %r13, %r14
        ...
```

cycle #	0	1	2	3	4	5	6	7	8
`cmpq %r8, %r9`	F	D	E	M	W				
`jne LABEL`		F	D	E	M	W			
`xorq %r10, %r11`			F	D	E	M	W		
`movq %r11, 0(%r12)`				F	D	E	M	W	
…									

jXX: speculating wrong

jXX: speculating wrong

	cycle # 0	1	2	3	4	5	6	7	8	
`cmpq %r8, %r9`	F	D	E	M	W					
`jne LABEL`		F	D	E	M	W				
`xorq %r10, %r11`			F	D						
`(inserted nop)`					E	M	W			
`movq %r11, 0(%r12)`				F						
`(inserted nop)`					D	E	M	W		
`LABEL: addq %r8, %r9`					F	D	E	M	W	
`imul %r13, %r14`						F	D	E	M	W
…										

jXX: speculating wrong

	cycle #	0	1	2	3	4	5	6	7	8	
`cmpq %r8, %r9`		F	D	E	M	W					
`jne LABEL`			F	D	E	M	W				
`xorq %r10, %r11`				F	$\boxed{\text{D}}$	instruction “squashed”					
`(inserted nop)`						E	M	W			
`movq %r11, 0(%r12)`					$\boxed{\text{F}}$	instruction “squashed”					
`(inserted nop)`						D	E	M	W		
`LABEL: addq %r8, %r9`						F	D	E	M	W	
`imul %r13, %r14`							F	D	E	M	W

…

“squashed” instructions

on misprediction need to undo partially executed instructions

mostly: remove from pipeline registers

more complicated pipelines: replace written values in cache/registers/etc.

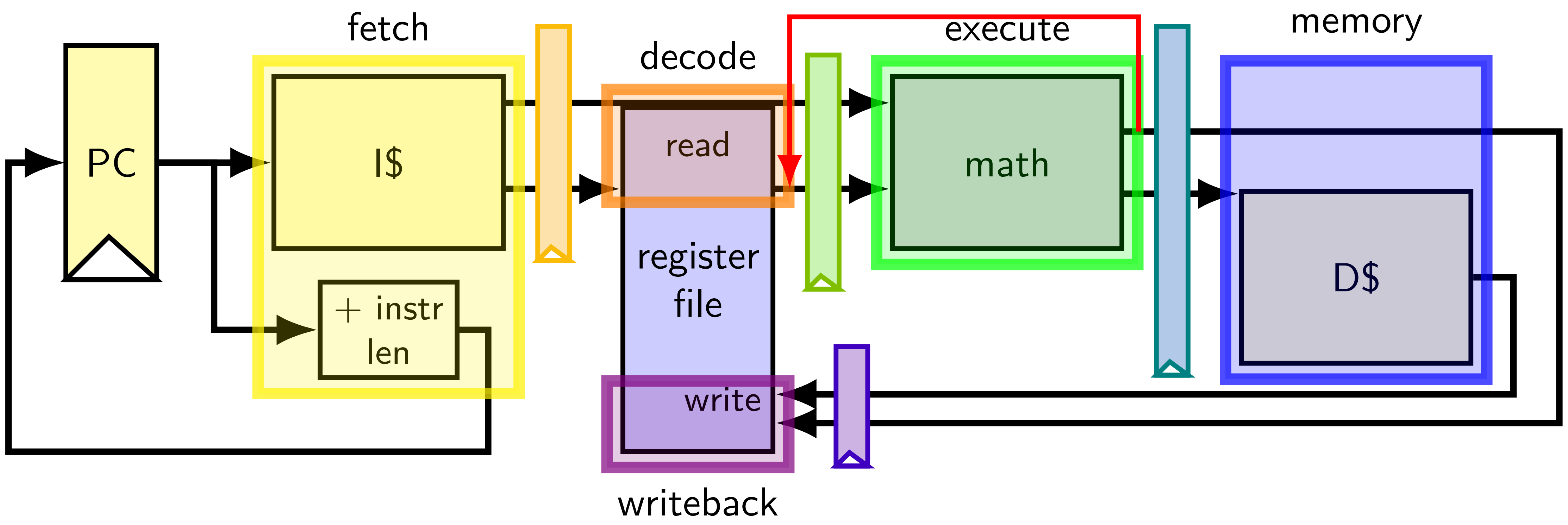
opportunity

```
// initially %r8 = 800,
//           %r9 = 900, etc.
0x0: addq %r8, %r9
0x2: addq %r9, %r8
...
```

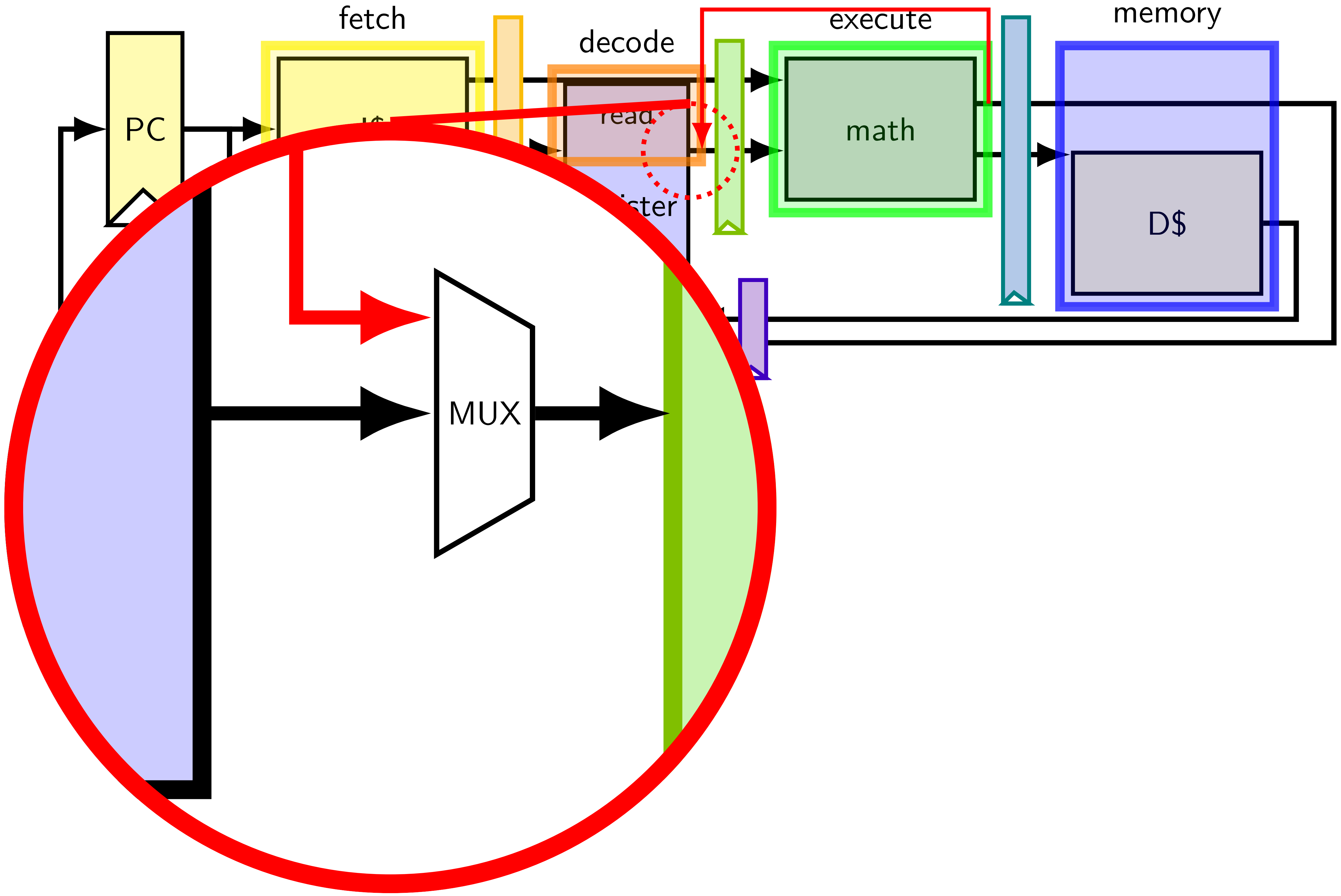
	fetch	fetch/decode		decode/execute			execute/memory		memory/writeback	
cycle	PC	rA	rB	R[rB	R[rB]	rB	sum	rB	sum	rB
0	0x0									
1	0x2	8	9							
2		9	8	800	900	9				
3				900	800	8	1700	9		
4							1700	8	1700	9
5									1700	8

should be 1700

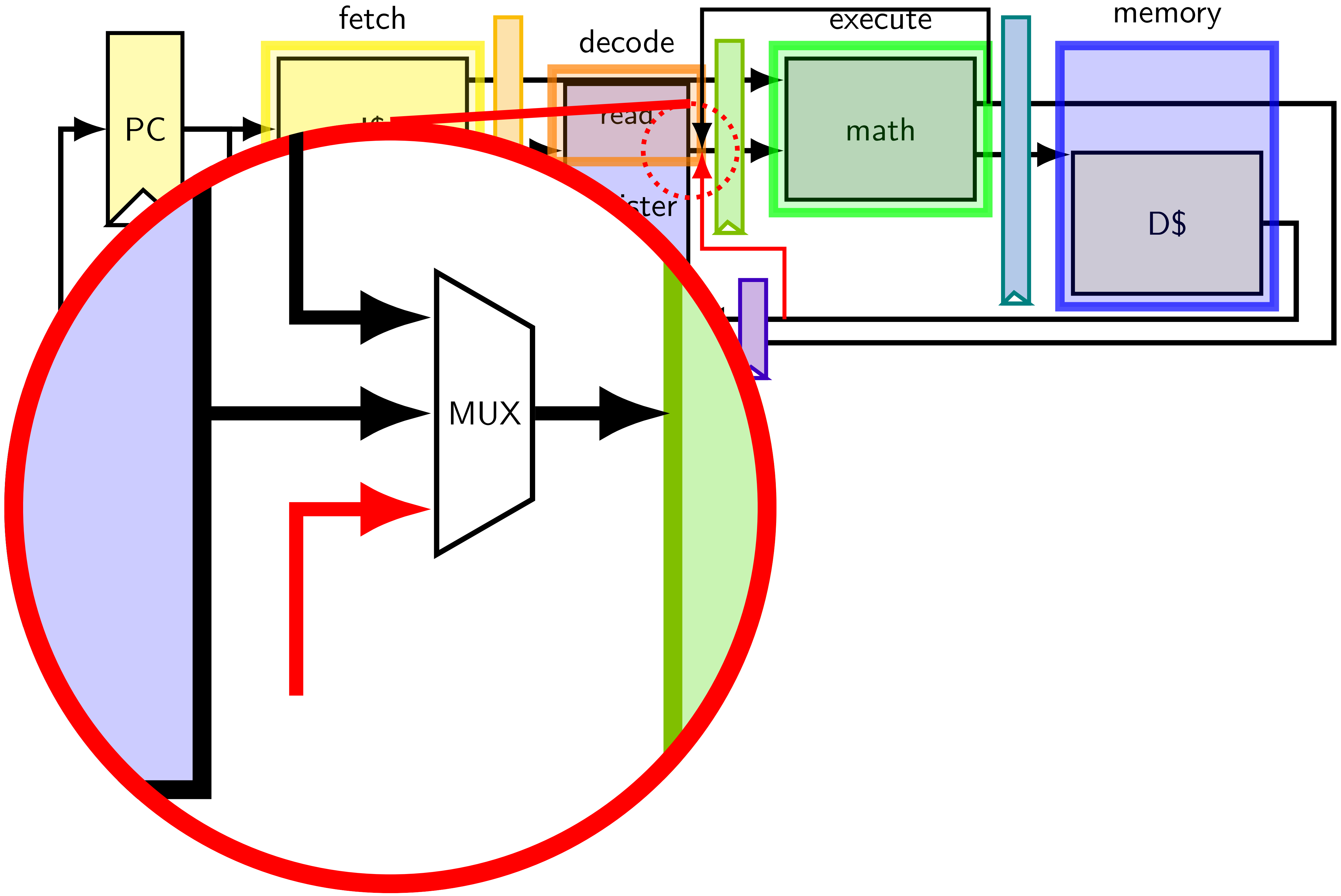
exploiting the opportunity



exploiting the opportunity



exploiting the opportunity



opportunity 2

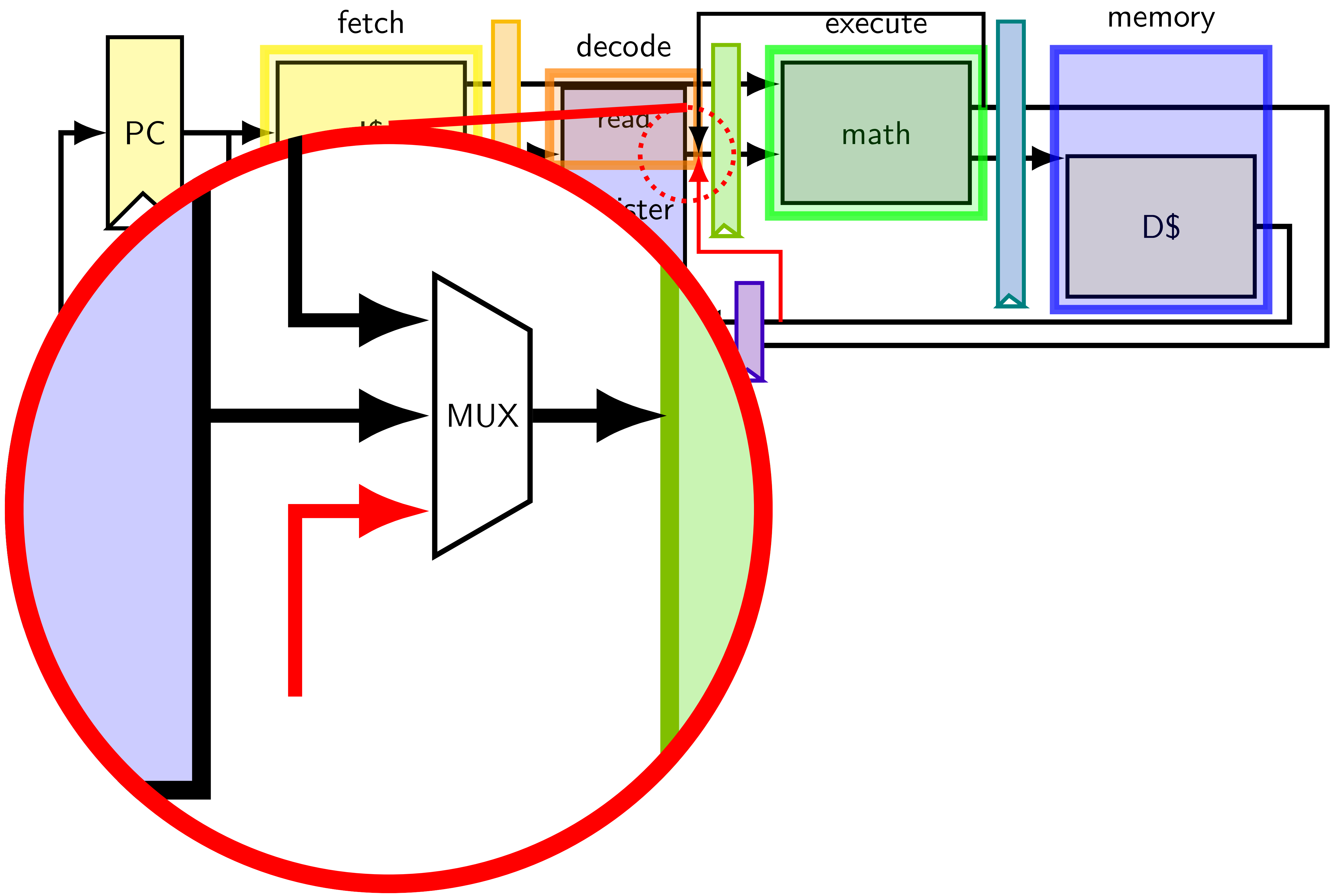
```
// initially %r8 = 800,
//           %r9 = 900, etc.
0x0: addq %r8, %r9
0x2: nop
0x3: addq %r9, %r8
...
```

	fetch	fetch/decode		decode/execute			execute/memory		memory/writeback	
cycle	PC	rA	rB	R[rB	R[rB]	rB	sum	rB	sum	rB
0	0x0									
1	0x2	8	9							
2	0x3	---	---	800	900	9				
3		9	8	---	---	---	1700	9		
4				900	800	8	---	---	1700	9
5							1700	9	---	---
6									1700	9

should be 1700

exploiting the opportunity

exploiting the opportunity



exercise: forwarding paths

cycle #	0	1	2	3	4	5	6	7	8
`addq %r8, %r9`	F	D	E	M	W				
`subq %r8, %r10`		F	D	E	M	W			
`xorq %r8, %r9`			F	D	E	M	W		
`andq %r9, %r8`				F	D	E	M	W	

in subq, %r8 is ____ addq.

in xorq, %r9 is ____ addq.

in andq, %r9 is ____ addq.

in andq, %r9 is ____ xorq.

- assume regfile cannot read value while being written
- A: not forwarded from
- B-D: forwarded to decode from {execute,memory,writeback} stage of

unsolved problem

```
movq 0(%rax), %rbx
subq %rbx, %rcx
```

cycle #	0	1	2	3	4	5	6	7	8
	F	D	E	M	W				
		F	D	E	M	W			

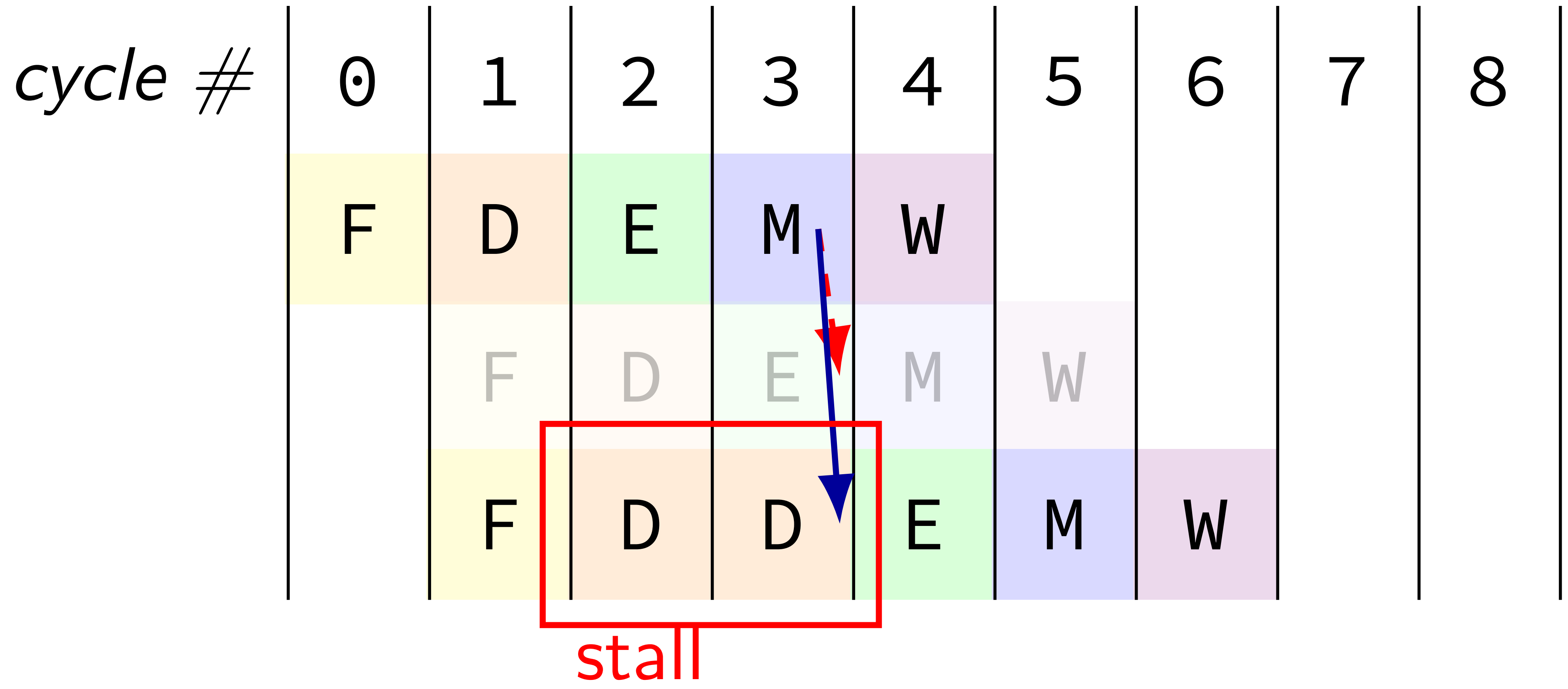
value from memory stage too late for forwarding alone

combine stalling and forwarding to resolve hazard

assumption in diagram: hazard detected in `subq`'s decode stage
  (since easier than detecting it in fetch stage)

unsolved problem

```
movq 0(%rax), %rbx
subq %rbx, %rcx
subq %rbx, %rcx
```



value from memory stage too late for forwarding alone

combine stalling and forwarding to resolve hazard

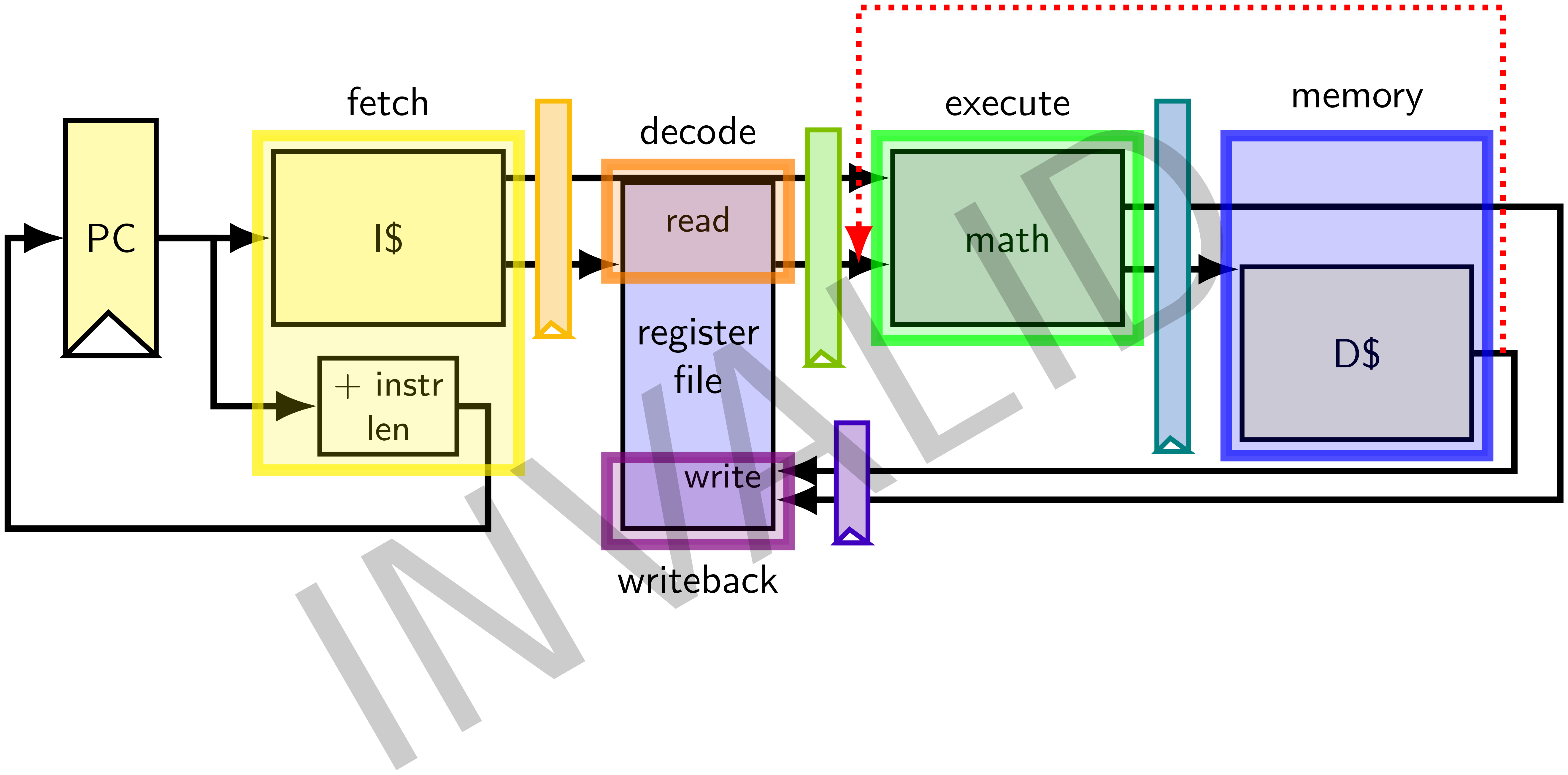
assumption in diagram: hazard detected in `subq`'s decode stage
(since easier than detecting it in fetch stage)

solveable problem

```
movq 0(%rax), %rbx
movq %rbx, 0(%rcx)
```

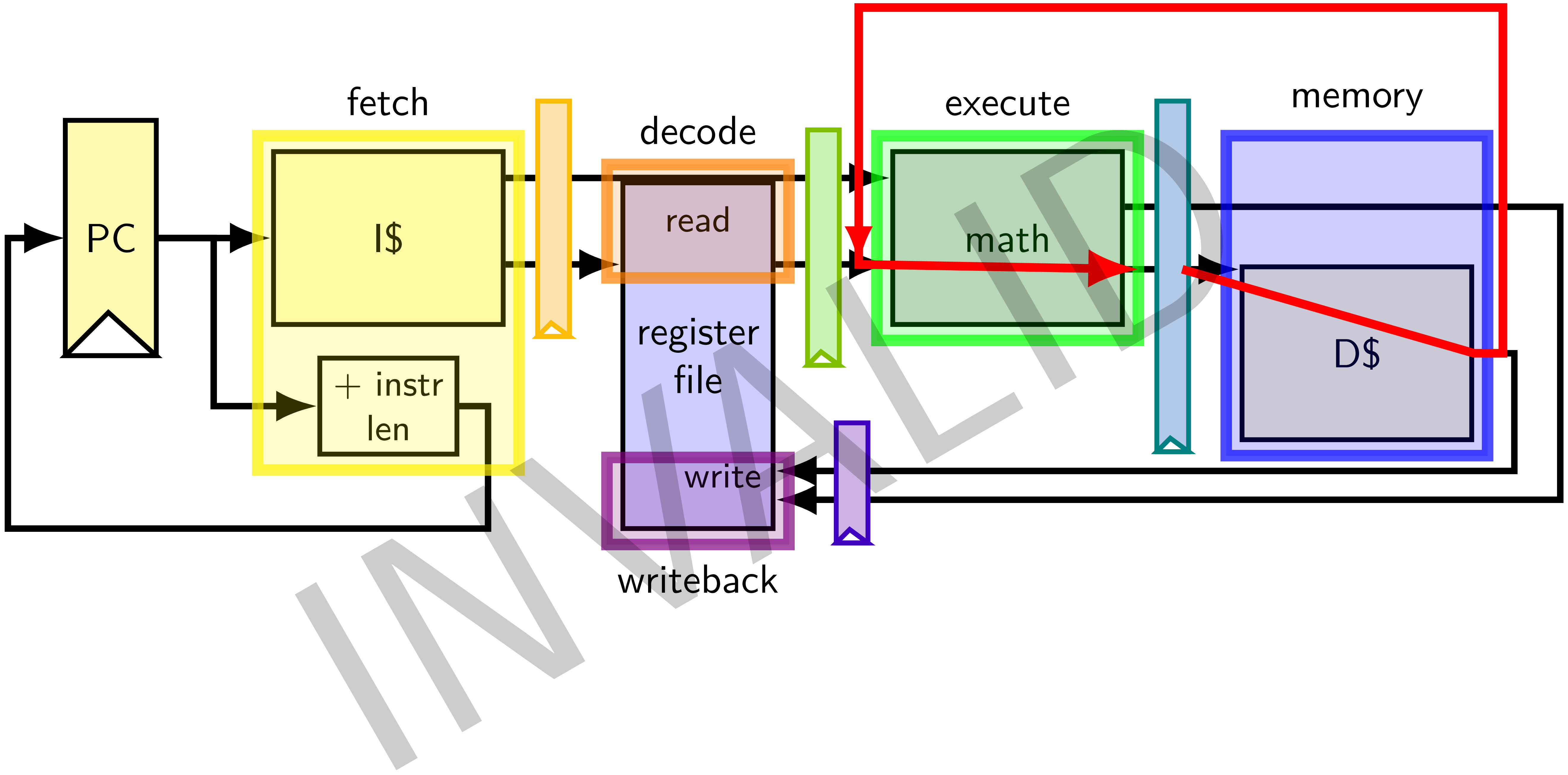
cycle #	0	1	2	3	4	5	6	7	8
	F	D	E	M	W				
		F	D	E	M	W			

why can't we...



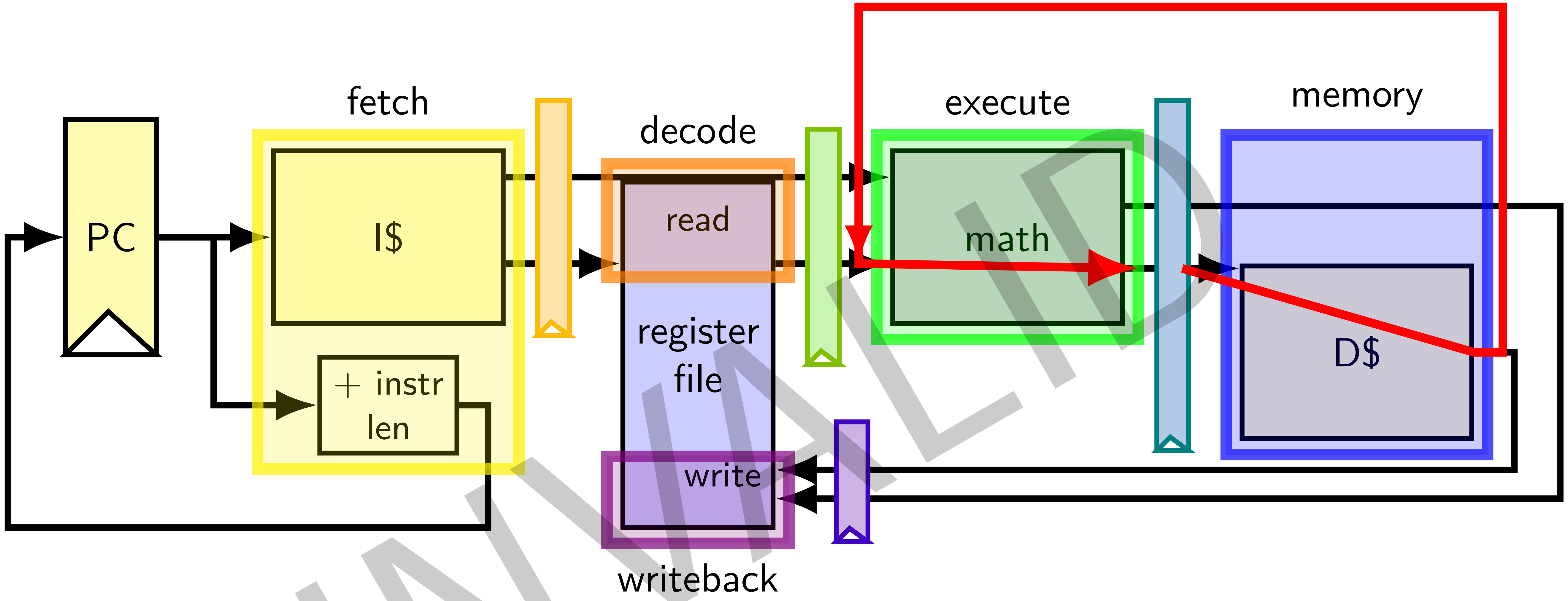
fetch
decode
execute
memory
PC
I$
+ instr
len
read
register
file
write
writeback
math
D$
INVALID

why can't we...



fetch
decode
execute
memory
PC
I$
+ instr
len
read
register
file
write
writeback
math
D$
INVALID

why can't we...



clock cycle needs to be long enough
to go through data cache AND
to go through math circuits!
(which we were trying to avoid by putting them in separate stages)

throughput exercise

exercise

suppose 5-stage pipeline
- using forwarding, stalling like we discussed

1 ns cycle time

5% are branches
- 60% correct predictions

1% are uses of value just after loading it from memory

assume negligible cache misses

estimated throughput?

exercise solution

97% 1 cycle (until next useful instruction fetched)

40% of 5% = 2% 3 cycles

- 2 “wasted” fetches for misprediction

1% 2 cycles

- 1 “wasted” fetch for data hazard stall

average cycles/instruction = 1$972$1% = 1.05

1 billion instructions / second $\div$ 1.05 cycles/instruction = 0.952 billion instr/sec

hazards versus dependencies

dependency — X needs result of instruction Y?

- has potential for being messed up by pipeline
- (since part of X may run before Y finishes)

hazard — will it not work in some pipeline?

- *before* extra work is done to "resolve" hazards
- multiple kinds: so far, *data hazards*, *control hazards*

ex.: dependencies and hazards (1)

ex.: dependencies and hazards (1)

```
addq    %rax,   %rbx
subq    %rax,   %rcx
movq    $100,   %rcx
addq    %rcx,   %r10
addq    %rbx,   %r10
```

where are dependencies?
which are hazards in our pipeline?
which are resolved with forwarding?

ex.: dependencies and hazards (1)

```
addq    %rax,   %rbx
subq    %rax,   %rcx
movq    $100,   %rcx
addq    %rcx,   %r10
addq    %rbx,   %r10
```

where are dependencies?
which are hazards in our pipeline?
which are resolved with forwarding?

ex.: dependencies and hazards (1)

```
addq    %rax,   %rbx
subq    %rax,   %rcx
movq    $100,   %rcx
addq    %rcx,   %r10
addq    %rbx,   %r10
```

where are dependencies?
which are hazards in our pipeline?
which are resolved with forwarding?

ex.: dependencies and hazards (1)

```
addq    %rax,   %rbx
subq    %rax,   %rcx
movq    $100,   %rcx
addq    %rcx,   %r10
addq    %rbx,   %r10
```

where are dependencies?
which are hazards in our pipeline?
which are resolved with forwarding?

pipeline with different hazards

example: 4-stage pipeline: fetch/decode/execute+memory/writeback

```
                 // 4 stage  // 5 stage
addq %rax, %r8   //          // W
subq %rax, %r9   // W        // M
xorq %rax, %r10  // EM       // E
andq %r8,  %r11  // D        // D
```

(assuming register file does not read while writing)

addq/andq is hazard with 5-stage pipeline

addq/andq is *not* a hazard with 4-stage pipeline

more hazards with more pipeline stages

pipeline with different hazards

example: 4-stage pipeline: fetch/decode/execute+memory/writeback

```
                    // 4 stage  // 5 stage
addq %rax, %r8      //          // W
subq %rax, %r9      // W        // M
xorq %rax, %r10     // EM       // E
andq %r8,  %r11     // D        // D
```

(assuming register file does not read while writing)

addq/andq is hazard with 5-stage pipeline

addq/andq is *not* a hazard with 4-stage pipeline

more hazards with more pipeline stages

pipeline with different hazards

example: 4-stage pipeline: fetch/decode/execute+memory/writeback

```
                  // 4 stage   // 5 stage
addq %rax, %r8    //           // W
subq %rax, %r9    // W         // M
xorq %rax, %r10   // EM        // E
andq %r8,  %r11   // D         // D
```

(assuming register file does not read while writing)

addq/andq is hazard with 5-stage pipeline

addq/andq is *not* a hazard with 4-stage pipeline

more hazards with more pipeline stages

exercise: different pipeline

split execute into two stages: F/D/E1/E2/M/W

result only available near end of second execute stage

where does forwarding, stalls occur?

	cycle #	0	1	2	3	4	5	6	7	8
(1) `addq %rcx, %r9`		F	D	E1	E2	M	W			
(2) `addq %r9, %rbx`										
(3) `addq %rax, %r9`										
(4) `movq %r9, 8(%rbx)`										
(5) `movq %rcx, %r9`										

exercise: different pipeline

split execute into two stages: F/D/E1/E2/M/W

cycle #	0	1	2	3	4	5	6	7	8
`addq %rcx, %r9`	F	D	E1	E2	M	W			
`addq %r9, %rbx`									
`addq %rax, %r9`									
`movq %r9, 8(%rbx)`									

exercise: different pipeline

split execute into two stages: F/D/E1/E2/M/W

cycle #	0	1	2	3	4	5	6	7	8
`addq %rcx, %r9`	F	D	E1	E2	M	W			
`addq %r9, %rbx`		F	D	E1	E2	M	W		
`addq %rax, %r9`								W	
`movq %r9, 8(%rbx)`				F	D	E1	E2	M	W

r9 not available yet — can't forward here
so try stalling in addq's decode…

exercise: different pipeline

split execute into two stages: F/D/E1/E2/M/W

cycle #	0	1	2	3	4	5	6	7	8	
`addq %rcx, %r9`	F	D	E1	E2	M	W				
`addq %r9, %rbx`		F	D	E1	E2	M	W			
`addq %r9, %rbx`		F	D	D	E1	E2	M	W		
`addq %rax, %r9`								W		
`addq %rax, %r9`			F	F	D	E1	E2	M	W	
`movq %r9, 8(%rbx)`				F	D	E1	E2	M	W	
`movq %r9, 8(%rbx)`					F	D	E1	E2	M	W

after stalling once, now we can forward

exercise: different pipeline

split execute into two stages: F/D/E1/E2/M/W

	cycle #	0	1	2	3	4	5	6	7	8	
`addq %rcx, %r9`		F	D	E1	E2	M	W				
`addq %r9, %rbx`			F	D	E1	E2	M	W			
`addq %r9, %rbx`			F	D	D	E1	E2	M	W		
`addq %rax, %r9`				F	D	E1	E2	M	W		
`addq %rax, %r9`				F	F	D	E1	E2	M	W	
`movq %r9, 8(%rbx)`					F	D	E1	E2	M	W	
`movq %r9, 8(%rbx)`						F	D	E1	E2	M	W

exercise: different pipeline

split execute into two stages: F/D/E1/E2/M/W

cycle #	0	1	2	3	4	5	6	7	8		
addq %rcx, %r9	F	D	E1	E2	M	W					
addq %r9, %rbx		F	D	E1	E2	M	W				
addq %r9, %rbx		F	D	D	E1	E2	M	W			
addq %rax, %r9			F	D	E1	E2	M	W			
addq %rax, %r9			F	F	D	E1	E2	M	W		
movq %r9, 8(%rbx)				F	D	E1	E2	M	W		
movq %r9, 8(%rbx)					F	D	E1	E2	M	W	
movq %rcx, %r9						F	D	E1	E2	M	W

Backup slides

exercise: control hazard timing+forwarding?

with F/D/E/M/W: what is fetched when? what is forwarded?

cycle #	0	1	2	3	4	5	6	7	8	9	10
(1) **addq** %rcx, %r9											
(2) **jne** foo (predicted taken, actually not)											
(3) **subq** %rax, %r9											
(4) **call** bar											
(5) bar: **pushq** %r9											

[solution]: control hazard timing+forwarding?

with F/D/E/M/W: what is fetched when? what is forwarded?

[solution]: control hazard timing+forwarding?

with F/D/E/M/W: what is fetched when? what is forwarded?

cycle #	0	1	2	3	4	5	6	7	8	9	10
(1) **addq** `%rcx, %r9`	F	D	E	M							
(2) **jne** `foo` (predicted taken, actually not)		F	D	E							
(2b) `foo: ...` (mispred.)			F	D							
(2c) `...` (mispred.)				F							
(3) **subq** `%rax, %r9`											
(4) **call** `bar`											
(5) `bar:` **pushq** `%r9`											

[solution]: control hazard timing+forwarding?

with F/D/E/M/W: what is fetched when? what is forwarded?

cycle #	0	1	2	3	4	5	6	7	8	9	10
(1) **addq** %rcx, %r9	F	D	E	M							
				pass flags							
(2) **jne** foo (predicted taken, actually not)		F	D	E							
(2b) foo: ... (mispred.)			F	D							
(2c) ... (mispred.)				F							
(3) **subq** %rax, %r9											
(4) **call** bar											
(5) bar: **pushq** %r9											

[solution]: control hazard timing+forwarding?

with F/D/E/M/W: what is fetched when? what is forwarded?

cycle #	0	1	2	3	4	5	6	7	8	9	10	
(1) **addq** %rcx, %r9	F	D	E	M	W							
				pass flags								
(2) **jne** foo (predicted taken, actually not)		F	D	E	M	W						
(2b) foo: ... (mispred.)			F	D	E	M	W					“squashed”
(2c) ... (mispred.)				F	D	E	M	W				
(3) **subq** %rax, %r9												
(4) **call** bar												
(5) bar: **pushq** %r9												

[solution]: control hazard timing+forwarding?

with F/D/E/M/W: what is fetched when? what is forwarded?

cycle #	0	1	2	3	4	5	6	7	8	9	10
(1) `addq %rcx, %r9`	F	D	E	M	W						
				pass flags							
(2) `jne foo` (predicted taken, actually not)		F	D	E	M	W					
(2b) `foo: ...` (mispred.)			F	D	E	M	W				
(2c) `...` (mispred.)				F	D	E	M	W			
					receive corrected PC						
(3) `subq %rax, %r9`					F						
(4) `call bar`											
(5) `bar: pushq %r9`											

[solution]: control hazard timing+forwarding?

with F/D/E/M/W: what is fetched when? what is forwarded?

cycle #	0	1	2	3	4	5	6	7	8	9	10
(1) **addq** %rcx, %r9	F	D	E	M	W						
(2) **jne** foo (predicted taken, actually not)		F	D	E	M	W					
(2b) foo: ... (mispred.)			F	D	E	M	W				
(2c) ... (mispred.)				F	D	E	M	W			
(3) **subq** %rax, %r9					F	D	E	M	W		
(4) **call** bar						F	D	E	M	W	
(5) bar: **pushq** %r9							F	D	E	M	W

pass flags

receive corrected PC

%r9

[solution]: control hazard timing+forwarding?

with F/D/E/M/W: what is fetched when? what is forwarded?

cycle #	0	1	2	3	4	5	6	7	8	9	10
(1) **addq** %rcx, %r9	F	D	E	M	W						
(2) **jne** foo (predicted taken, actually not)		F	D	E	M	W					
(2b) foo: ... (mispred.)			F	D	E	M	W				
(2c) ... (mispred.)				F	D	E	M	W			
(3) **subq** %rax, %r9					F	D	E	M	W		
(4) **call** bar						F	D	E	M	W	
(5) bar: **pushq** %r9							F	D	E	M	W

pass flags

receive corrected PC

%r9 and %rsp

exercise: with different pipeline

with F/D/E1/E2/M/W

	cycle #	0	1	2	3	4	5	6	7	8	9	10
(1) `addq %rcx, %r9`												
(2) `jne foo` (not taken)												
(3) `subq %rax, %r9`												
(4) `call bar`												
(5) `bar: pushq %r9`												

[solution]: with different pipeline

with F/D/E1/E2/M/W

	cycle # 0	1	2	3	4	5	6	7	8	9	10			
(1) **addq** %rcx, %r9	F	D	E1	E2	M	W								
(2) **jne** foo (not taken)		F	D	E1	E2	M	W							
(2b) mispredicted			F	D	E1	E2	M	W						
(2c) mispredicted				F	D	E1	E2	M	W					
(2d) mispredicted					F	D	E1	E2	M	W				
(3) **subq** %rax, %r9						F	D	E1	E2	M	W			
(4) **call** bar							F	D	E1	E2	M	W		
(5) bar: **pushq** %r9								F	D*	D	E1	E2	M	W

exercise: forwarding paths (2)

cycle # `0 1 2 3 4 5 6 7 8`

```
addq %r8, %r9

subq %r8, %r9

ret (goes to andq)

andq %r10, %r9
```

in subq, %r8 is ____ addq.

in subq, %r9 is ____ addq.

in andq, %r9 is ____ subq.

in andq, %r9 is ____ addq.

A: not forwarded from

B-D: forwarded to decode from {execute,memory,writeback} stage of

exercise: predict+forward (1)

	cycle # 0	1	2	3	4	5	6	7	8
`addq %r8, %r9`	F	D	E	M	W				
`subq %r7, %r8`		F	D	E	M	W			
`jle foo (taken)`			F	D	E	M	W		
…									
…									
…									
`foo: andq %r9, %r8`									

if jle is *correctly predicted*:

in andq, %r9 is _____ addq.

in andq, %r8 is _____ subq.

A: not forwarded from [assume read while writing requires forwarding]

B-D: forwarded to decode from {execute,memory,writeback} stage of

exercise: predict+forward (2)

	cycle # 0	1	2	3	4	5	6	7	8
`addq %r8, %r9`	F	D	E	M	W				
`subq %r7, %r8`		F	D	E	M	W			
`jle foo (taken)`			F	D	E	M	W		
...									
...									
...									
`foo: andq %r9, %r8`									

if jle is *mispredicted* + resolved after jle's execute:

in andq, %r9 is ____ addq.

in andq, %r9 is ___ subq.

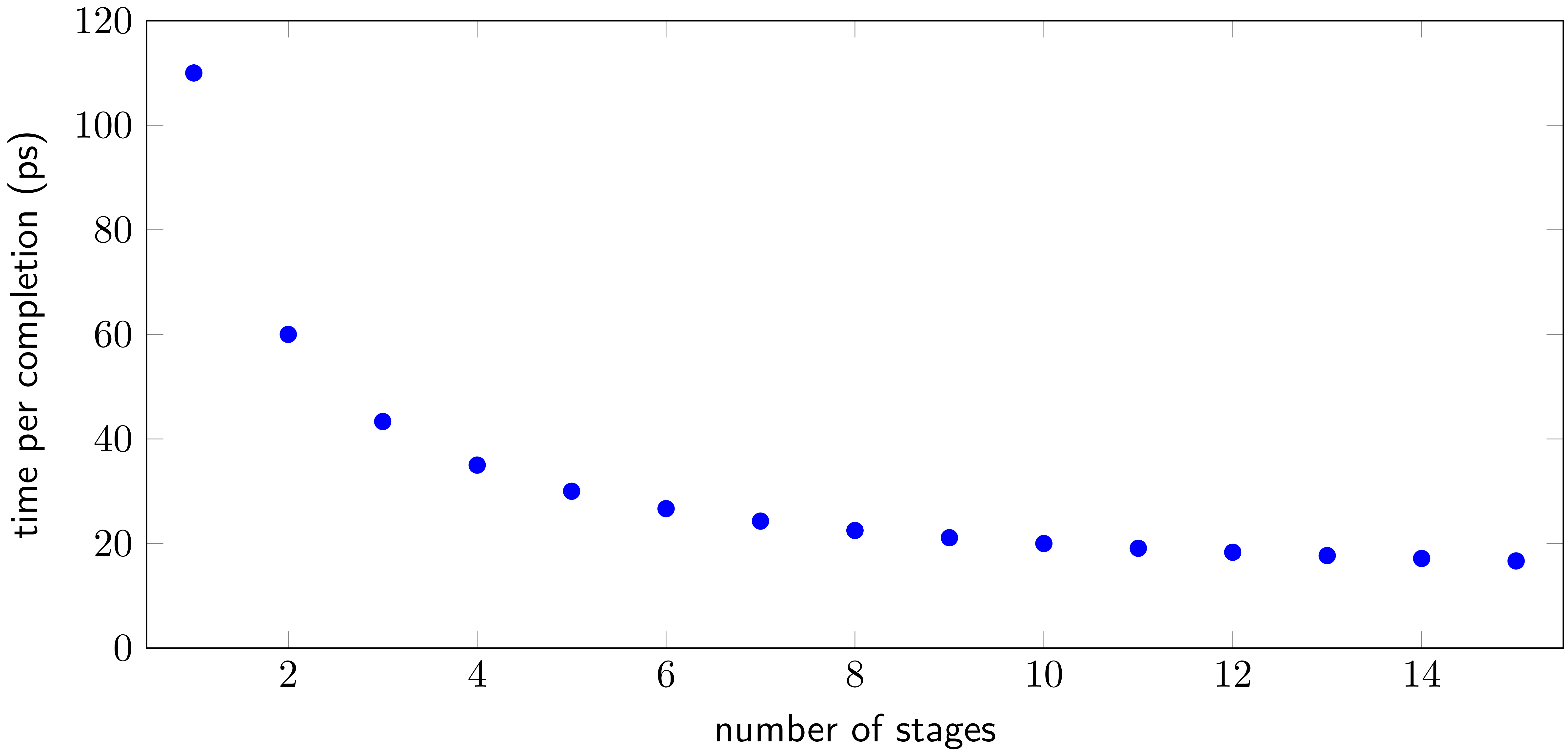
A: not forwarded from [assume read while writing requires forwarding]

B-D: forwarded to decode from {execute,memory,writeback} stage of

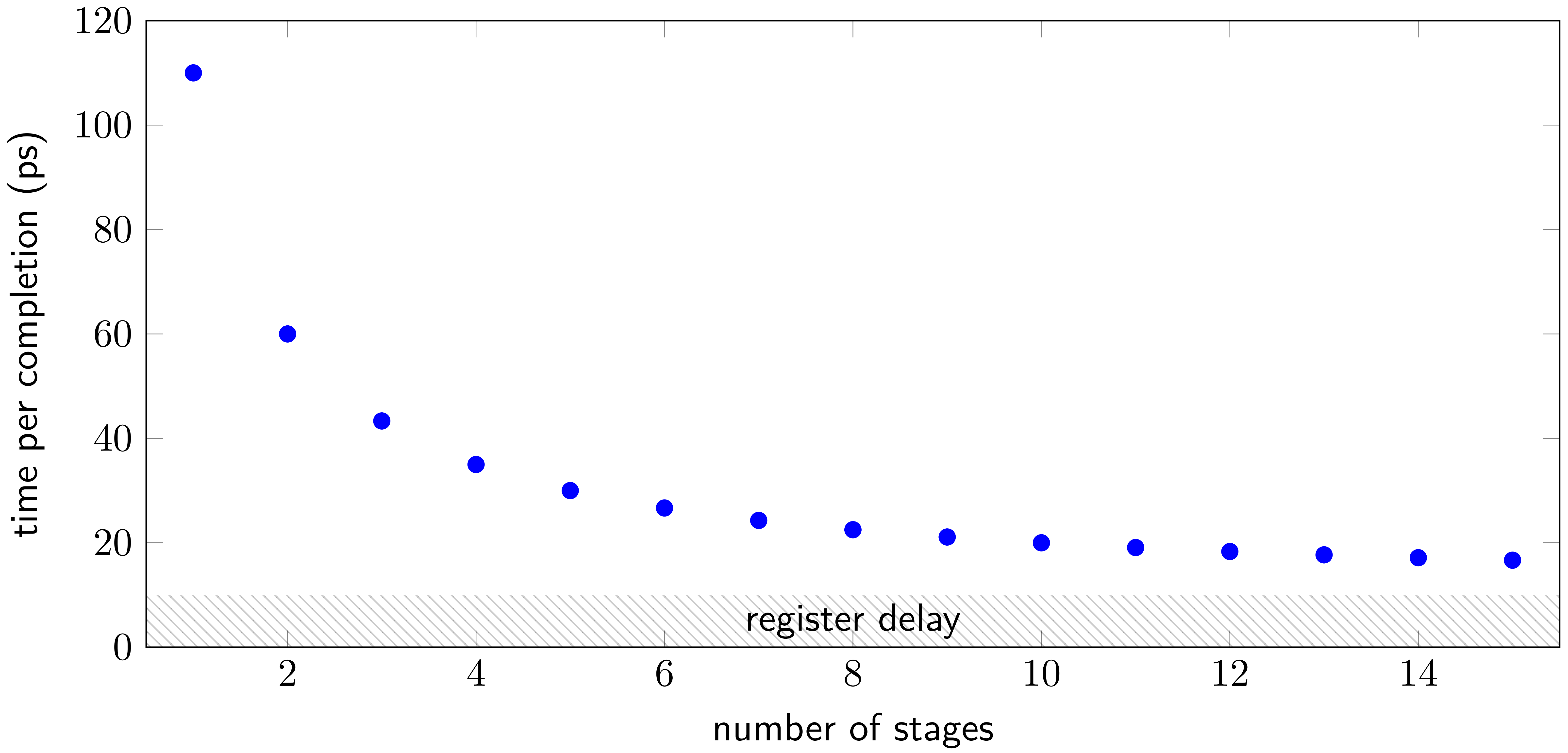
cycle #	0	1	2	3	4	5	6	7	8	
addq %r8, %r9	F	D	E	M	W					
subq %r7, %r8		F	D	E	M	W				
jle foo (taken)			F	D	E	M	W			
(mispredicted)				F	D					
(mispredicted)					F					
…										
foo: **andq** %r9, %r8						F	D	E	M	W

diminishing returns: register delays

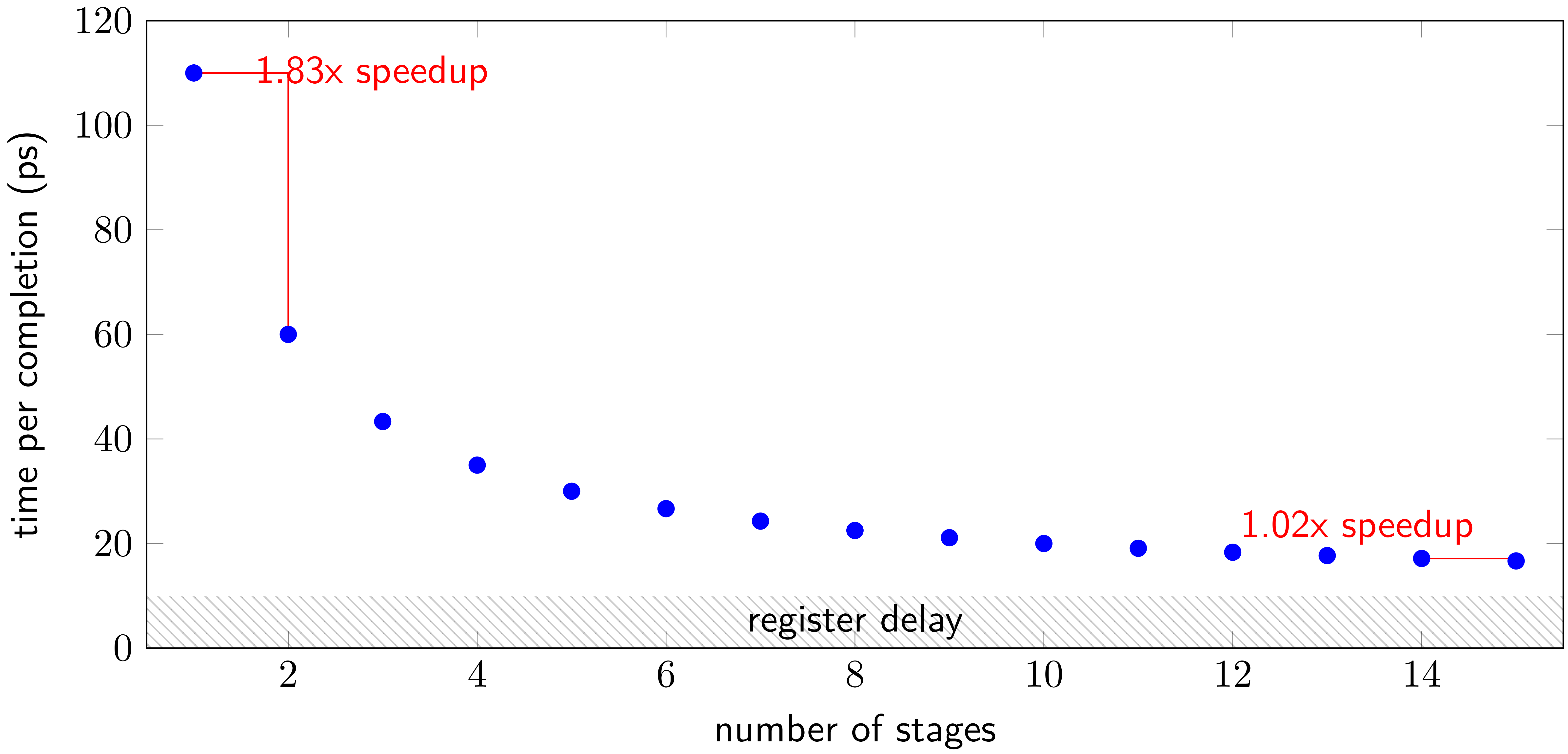
diminishing returns: register delays



diminishing returns: register delays

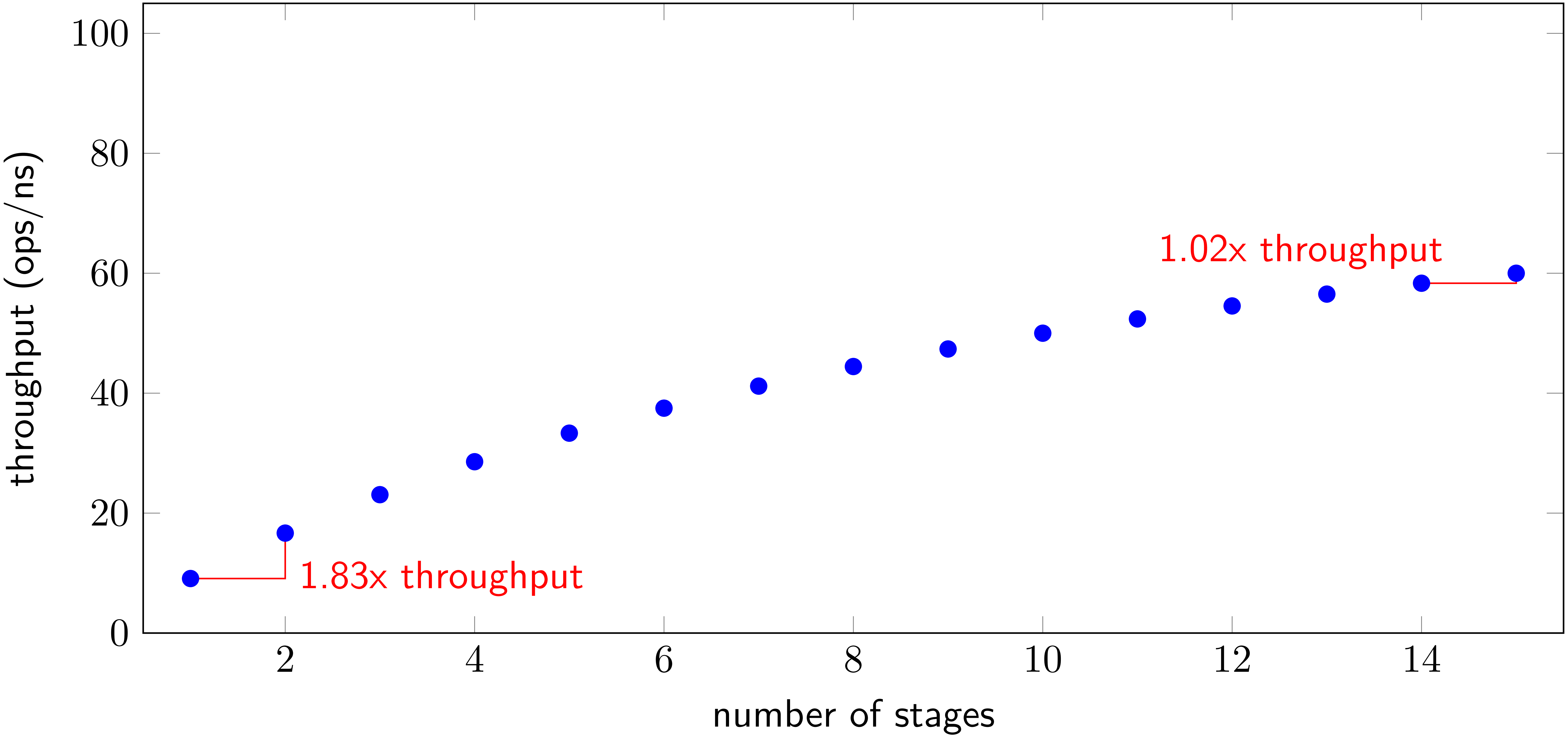


diminishing returns: register delays

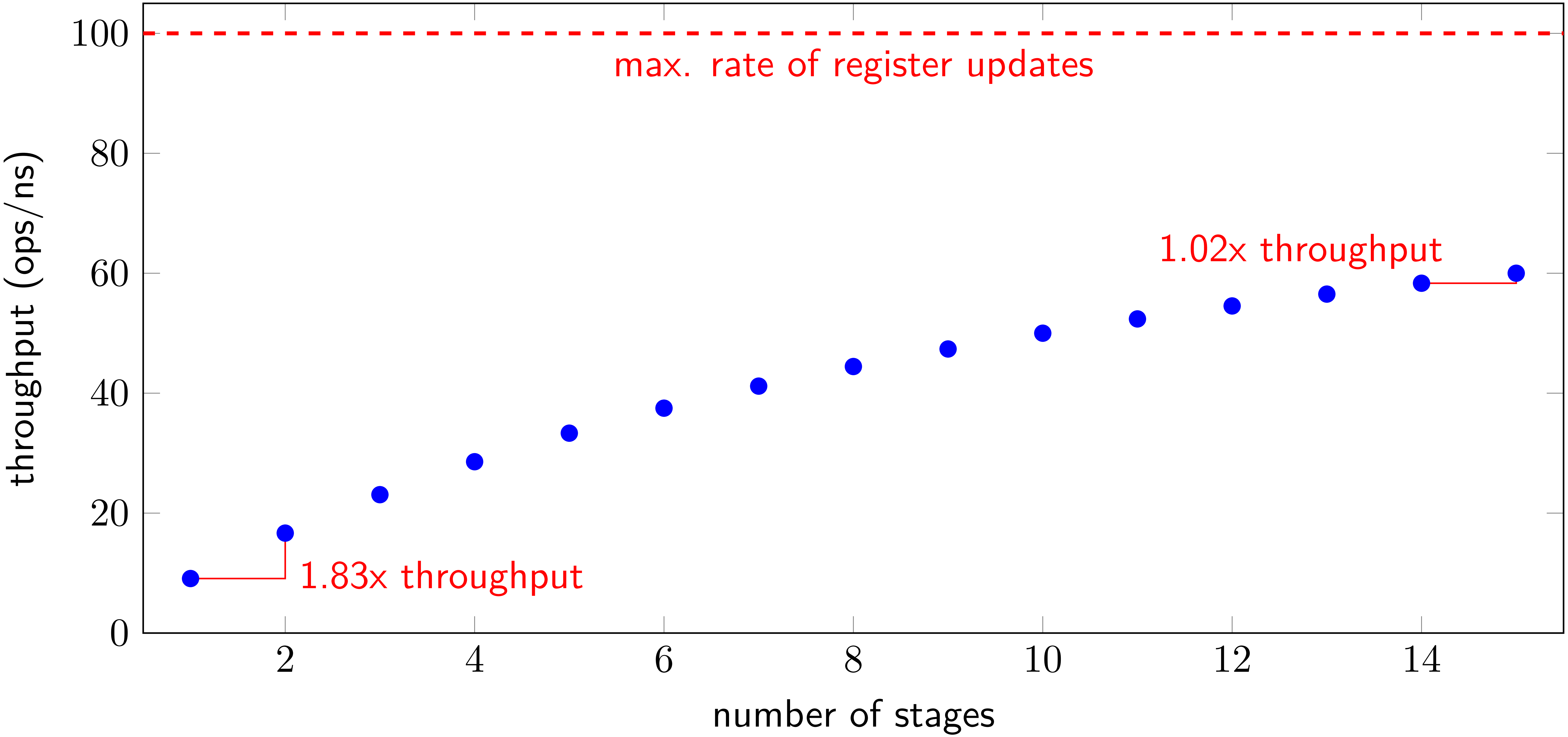


diminishing returns: register delays

diminishing returns: register delays



diminishing returns: register delays



importance of prediction

5-stage pipeline, predict not-taken versus always stall

* — ignoring data hazards

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hypothetical instruction mix

kind	portion	cycles (predict not-taken)	cycles (stall, no predict)
taken jXX	3%	3	3
non-taken jXX	5%	1	3
others	92%	1*	1*

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predict: $3 \times .03 + 1 \times .05 + 1 \times .92 = 1.06$ cycles/instr.

stall: $3 \times .03 + 3 \times .05 + 1 \times .92 = 1.16$ cycles/instr.

$(1.16 \div 1.06 \approx 1.09$x faster)

* — ignoring data hazards

prediction and OOO

deeper pipeline — much higher misprediction penalty